Preliminary Design Report

Imperial College Hyperloop

This report provides an overview on the progress made since October 2020 on **ICLoop**, the main project of the **Imperial College Vehicle Design Society**, to compete in the **SpaceX Pod competition**. This report outlines the research carried out by the team, the design choices made and some early prototype designs.

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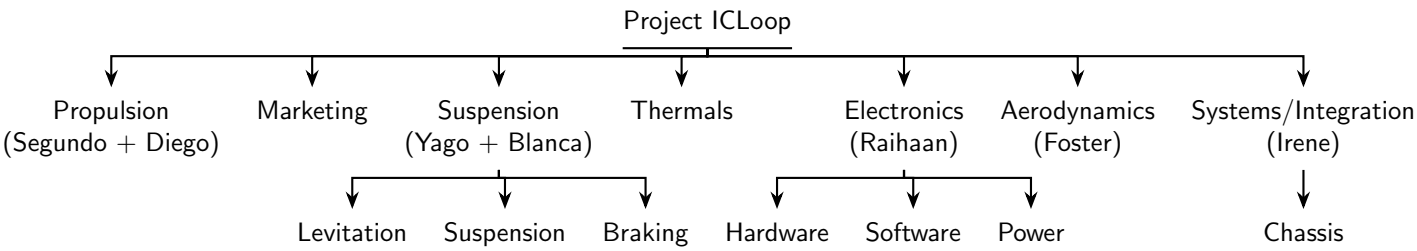
1 Introduction

The Hyperloop concept as it is known today, was introduced back in 2013 by Elon Musk in a white paper. The initial concept, Hyperloop Alpha, was basic, not scientifically based and open to interpretation; today, the Virgin Hyperloop One is fully-functional system, breaking records and a testament to the extensive research and development made since. Very efficient linear electric motors were developed; magnetic levitation was achieved without the need for super cooling. The Imperial College Vehicle Design Society's Hyperloop project, ICLoop, aims to compete in the SpaceX 2022 Hyperloop Pod Competition. Primarily, the target is to design a Pod capable of reaching a peak velocity of 400 km/h, or to exceed it.

2 Team Structure

The ICLoop team is categorised into five sub-teams, each of which is made up of students from various departments at Imperial College, including Aeronautics, Electrical and Electronic Engineering, Materials, Mechanical Engineering, etc. These sub-teams are Aerodynamics, Propulsion, Suspension and Braking, Electronics and Software, and Chassis and Integration. This report is therefore divided into these categories.



3 Propulsion

3.1 Introduction

The main factor holding back the development of a full-scale Hyperloop is the development of a powerful and at the same time efficient propulsion system. The Propulsion sub-team has been given the task of developing a fully functional propulsion system that meets certain performance parameters while abiding by competition regulations.

The topics covered in the section are the decision making process, theory related to the technology being used, design process and testing and a final subsection addressing optimisation via machine learning. These subsections are not densely packed with theory, they only aim to display the team's knowledge of the underlying theory and the ways in which the propulsion system will be designed.

3.2 Objectives and Limitations

The main objective of the Imperial College Hyperloop (ICLoop) is to enter the 2022 SpaceX Pod competition, due to the cancellation of the 2021 Pod competition. Aside of this main goal, the propulsion sub-team has set an internal objective to have a measure of how successful the propulsion system on its own was. As has been previously established, one of the performance objectives of the pod is to reach a maximum speed of 400 km/h. In order to do this in the track provided for the SpaceX competition, we have calculated (see Appendix B) that the propulsion system will need to produce a thrust of at least 2000 N. It must be noted that the maximum velocity of the pod will not be dependant only on the thrust output, but also on the efficacy of the rest of the subsystems.

Since ICLoop will be entering a competition with an already existing track, design constraints are added. Due to the fixed track geometry the propulsion system will be designed and built around an already existing track (see Appendix A).

Further constraints were introduced by the need for the Levitation and Braking subsystem and the Propulsion subsystem to be perfectly compatible, since the performance of these two systems is highly co-dependent. For example, if a Linear Induction Motor (LIM) is to be used as the propulsion system, the levitation and braking systems would most likely have to be contactless systems. Although a wheeled braking and suspension system could be paired with a LIM, since the advantage of using a LIM is having a contactless system with no moving parts, the only viable pairing for a LIM propulsion system is a contactless levitation and braking system (discussed further in chapters 4 and 6).

3.3 Viable options

The propulsion sub-team had a wide range of propulsion methods and configurations to choose from. The main propulsion systems for our specific application were rotational electric motor, linear induction motor (LIM), and cold gas thrusters. The upsides and downsides of all the systems considered will be discussed in this section, together with their compatible levitation and suspension systems

Rotational motors

The rotational electric motor is the most widely used and known type of electric motor. It consists of two main components - the stator and the rotor. As per its name, the stator is the stationary part of the motor and it consists of the frame, core, and winding. The purpose of the stator within the motor is to generate the magnetic field. The rotor is the rotating part of the motor and it consists of the core and winding. The benefit of using these types of motors is that they have high efficiencies [15] and also the rotational speed can be easily adjusted by changing the frequency of the alternating current electricity

supply. However, they work by connecting to an axle which in turn is connected to a wheel. This means that there will be friction between the wheels and the surface, creating an extra opposing force to the pod's direction of travel. Rotational motors in Hyperloop pods need a contact suspension method since the nature of the thrust being generated is due to the contact between the wheel being driven by the motor and the track.

Linear Induction Motor

The LIM is similar to the rotational electric motor in that it consists of the same two main components: stator and rotor. The difference between the two is that LIM does not produce torque but instead, it produces a linear force to propel the Pod forward. Moreover, this allows for it to be contactless and avoid any friction due to surfaces in contact [16]. If the stator is in the track, the rotor will be on the Pod, and viceversa. If the stator is on the ground it means that the energy source for the motor can also be off the vehicle; saving vital weight and making it easier to accelerate. Despite this seeming like a promising approach it is not possible in the context of the SpaceX competition, since the track has already been built and does not have a long primary configuration (stator embedded in the track). This configuration does not seem very scalable, since a country-wide track network where electromagnets must be placed along all the length of the track would be too expensive to build.

A variation of the LIM is the double sided LIM, as opposed to the basic single-sided LIM (SLIM). The advantage of this variation is that it is easier for the vehicle to be in line with the centre of the track, preventing adverse effects.

All types of Linear Induction Motors are affected by end-effects [17]. This is because the motors are not infinitely long thus the magnetic fields at the end have to curl around â producing unfavourable conditions for the Pod. This reduces the total thrust output linear induction motors, making them less energy efficient than their rotational counterparts. It was previously mentioned that a rotational motor requires a contact suspension in order to maintain the driving wheel(s) in contact with the track, but since the LIM uses contactless technology a levitation system can be introduced to completely avoid contact friction and reduce even further heating effects and friction. This means the LIM system would be compatible with both a traditional wheel system and a levitation system.

Cold Gas Thrusters

Cold gas thrusters use the expansion of a pressurised gas to generate thrust. Cold gas thrusters utilise inert gasses to propel the vehicle forward [16]. However, this means the gas has to be stored on board the vehicle â adding extra weight and taking up more space. A very significant flaw of this system for this Hyperloop application is that if this inert gas is released into the pressured track, it will increase the pressure of the track, forcing the pumps to be used again to depressurise the entire track, which would prove very costly. This propulsion approach can use both a contact suspension or levitation technology since the nature of this propulsion does not depend on interactions with the track.

3.3.1 Final decision

The propulsion system to be used in the Pod is a double sided LIM. It has been deemed to be the best propulsion system due to its lack of moving parts, high acceleration and great scalability properties. While cold gas thrusters were rapidly dismissed, using a rotational motor was very seriously considered but eventually a LIM was chosen. A LIM was preferred over all other propulsion systems in large part because it is contactless and allows for frictional forces between the pod and track to be minimal.

Other attractive qualities of the LIM (long stator) propulsion system are how easy it is to scale it up from a small competition pod to a continent-wide transportation network, the fact that it allows for the pod to not be in contact with the track and its lack of moving parts.

A question was posed by the society committee, which asked whether the objective of the project was to

simply win the SpaceX Competition or to, in doing so, contribute as much as possible to the development of Hyperloop technology. The team concluded that since the competition will be held in two years, this will be enough time to properly develop a functioning LIM and in doing so attempt to further advance Hyperloop technology past building a wheeled supercar that fits into the depressurised track.

3.4 LIM Theory

3.4.1 Motor Basics

To understand how a Linear Induction Motor (LIM) works, it is always easier to begin with a rotational motor. A rotational motor mainly consists of a stator and a rotor. The stator is stationary whilst the rotor rotates inside it, creating the mechanical motion required. A diagram of a rotational motor is seen below.

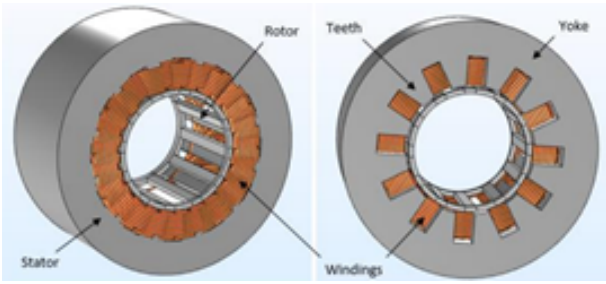


Figure 1: Diagram of a rotational motor

Assuming the stator consists of teeth around which windings are coiled, the current passing through the windings will induce a magnetic field. Using a magnetically permeable material such as iron will allow to direct the induced magnetic field through the material (i.e. the teeth). The alternating current in the windings coiled around the statorâs teeth will change the direction of the magnetic field. Thus, the stator creates a changing magnetic field depending on the direction of the current passing through each tooth as seen below in the squirrel cage example.

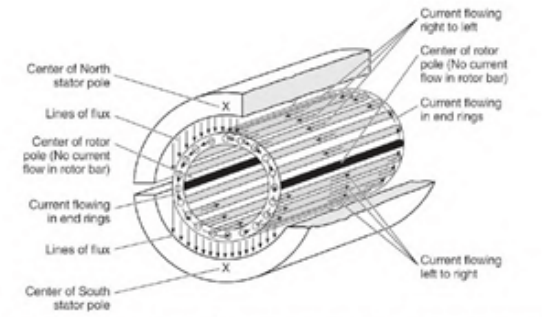


Figure 2: Diagram of a squirrel cage [2]

The changing magnetic field will induce the eddy currents inside the rotor, creating a rotational force (torque) following the right-hand rule and causing the rotor to rotate. This force has a direction tangent to the circumference of the squirrel cage.

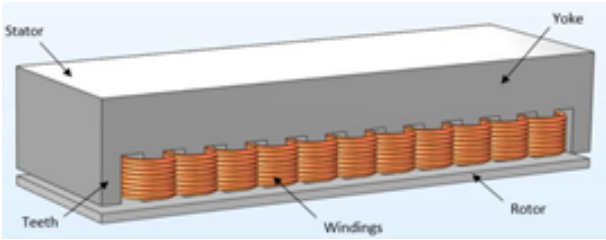


Figure 3: LIM consisting of the stator (on pod) and rotor(rail)

The Linear Induction Motor can be thought of as a rolled out rotational motor, as seen above. The windings will continue to carry an alternating current around the teeth. The stator's current will once more induce a magnetic field which will generate a perpendicular force to stator's teeth. Since the stator has a linear shape, rather than generating a rotational force as in the rotational motor's case, the magnetic field will generate a linear thrust parallel to the stator's linearity. This means the current in adjacent teeth windings must be out of phase to keep the LIM constantly moving. This is called the polyphase excitation and will be discussed in more detail in the following section.

An advantage of a LIM is that no additional contact mechanism is required to translate the thrust generated unlike in a rotational motor. This reduces the podâs total mass, allowing for greater speeds. In addition, since a LIM propulsion does not depend on adherence, its thrust will not be significantly affected by external conditions.

Disadvantages of a LIM compared to a rotational motor are that the LIM being of finite length will lose flux in front of and behind the Pod This results in a reduction of the thrust generated [3]. However, those flux leakages may be reduced by using a double-sided LIM, as seen below.

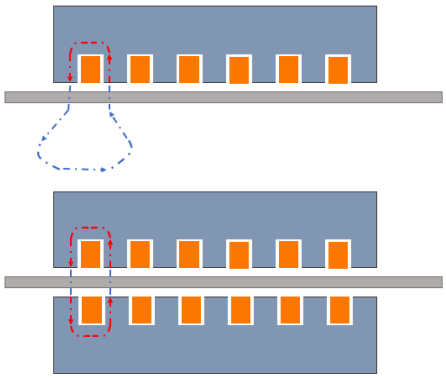


Figure 4: Magnetic flux in a double-sided LIM

Another disadvantage is that the rotor (rail) does not have gaps like the squirrel cage rotor, meaning it will be harder for the magnetic field to interact it. This will result in the squirrel cage rotor being slightly more efficient than the plate rotor. Since the rails used will be those designed by SpaceX, no gaps may be added to the rail (rotor). Unlike the rotational motor, the LIM will have a stationary rotor and a moving stator due to the track configuration.

3.4.2 Polyphase Excitation

The polyphase induction motor is arguably one of the most important motor types used in industry. In the LIM system we swap the rotational factors for magnetic fields and torque generation for linear fields; the basics for polyphase excitation remains the same.

At its core, polyphase excitation is the splitting up of an input into multiple constituent inputs with

varying phase but equal amplitude. The reason for this splitting becomes apparent. As the excitation of any motor, linear or rotational is proportional to the amplitude of the input voltage, in a sinusoid, there will be points of maximum excitation(the peaks), followed by a drop off where the amplitude is reducing - 'breaks' where it is not contributing as much. This would be terrible for the efficiency of the motor system, as the input voltage is only transferring torque for a fraction of the time. To increase efficiency we would aim to reduce the duration of the 'breaks' as much as possible.

Thus, splitting the input into multiple different waveforms with a phase difference with even phase shift will mean the maximum output torque(equivalent) is applied for as much time as possible. When you apply many of these polyphase waves simultaneously, a travelling field is produced, which follows the maximum points of the various waves of different phases. For imagination purposes, it is similar to a domino effect, where it follows the peaks of the out of phase waves.

It is common to have 3 unique phase inputs, each seperated by a phase shift of 120 degrees. For example:

- Acos(wt)
- Acos(wt + 120)
- Acos(wt + 240)

Having as many different 'polyphase' waveforms as possible is important for increasing efficiency and reducing breaks. This does, of course, come with the trade off of increasing complexity.

3.4.3 Travelling Stator Field

The magnetic field is created by the current passing through stator windings, according to Ampere's law. The magnetic field can be quantified by taking a path around a single coil, of a single phase.

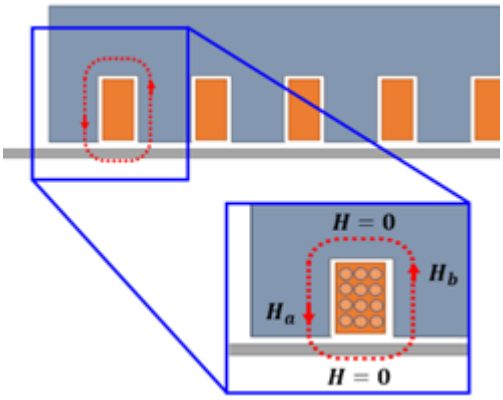


Figure 5: Closed path of the magnetic stator field around a coil of the stator [8]

For the path around the winding, we can obtain the following relation:

$$H_a \cdot g - H_b \cdot g = N \cdot I \tag{1}$$

where H is the magnetic field strength, g is the air gap distance, N is the number of windings and I the current passing through the windings. Therefore, since $H_a = -H_b$, then $\rightarrow H = NI/2g$.
The magnetic stator field can be divided into time and space varying components:

$$\sin(wt)$$

$$\sin\left(\frac{\pi \cdot y}{\tau_p}\right)$$

where y is the linear direction in which the motor translates when it generates thrust and τ_p is the pole pitch.
The equation therefore becomes:

$$H(y,t) = \frac{4}{\pi} \cdot \frac{N \cdot I}{2 \cdot g} \cdot \sin\left(\frac{\pi \cdot y}{\tau_p}\right) \cdot \sin(wt)) \tag{2}$$

Now, considering a 3 phase input of equal amplitude and with a phase shift of $\frac{2\pi}{3}$:

$$H_1(y,t) = \frac{4}{\pi} \cdot \frac{N \cdot I}{2 \cdot g} \cdot \sin\left(\frac{\pi \cdot y}{\tau_p}\right) \cdot \sin(wt) \tag{3}$$

$$H_2(y,t) = \frac{4}{\pi} \cdot \frac{N \cdot I}{2 \cdot g} \cdot \sin\left(\frac{\pi \cdot y}{\tau_p} + \frac{2\pi}{3}\right) \cdot \sin\left(wt + \frac{2\pi}{3}\right) \tag{4}$$

$$H_3(y,t) = \frac{4}{\pi} \cdot \frac{N \cdot I}{2 \cdot g} \cdot \sin\left(\frac{\pi \cdot y}{\tau_p} - \frac{2\pi}{3}\right) \cdot \sin\left(wt - \frac{2\pi}{3}\right) \tag{5}$$

By then taking into account the amplitude of each equation and superposing them the following expression is achieved:

$$H_{tot} = \frac{6}{\pi} \cdot \frac{NI}{2g} \cdot \cos\left(\frac{\pi \cdot y}{\tau_p} - wt\right) \tag{6}$$

This is achieved mathematically by converting all sine components of the previous equations into cosines and then summing them. This resulting superposition and the original equations can be seen in the following figure:

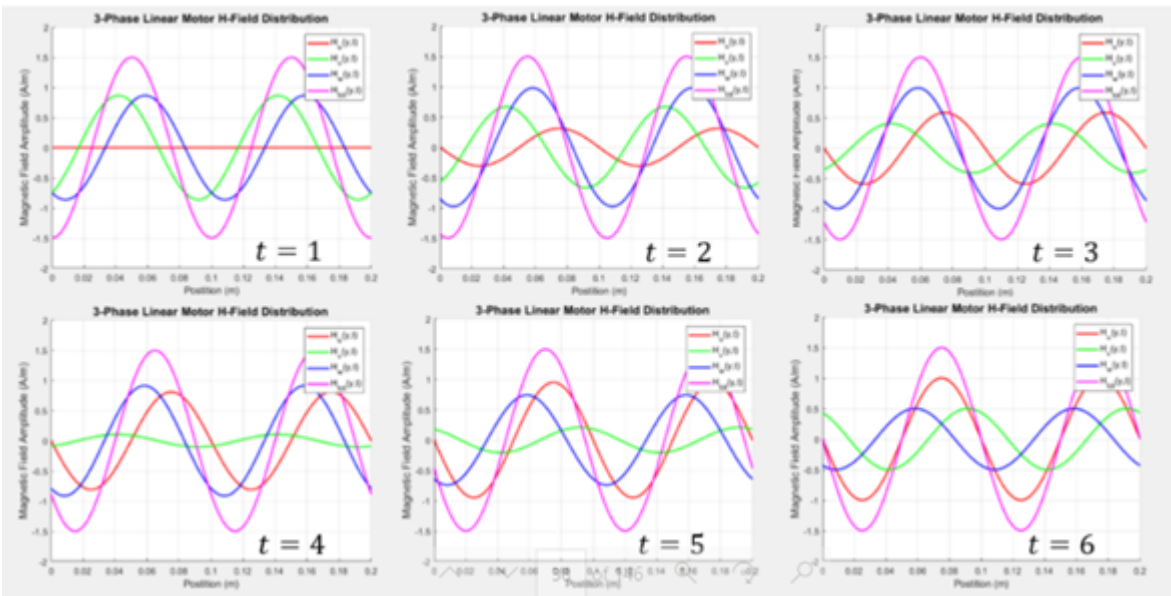


Figure 6: Progression in time of the 3-Phase Stator Fields and Total Stator Field [8]

The only field to move horizontally in position is the total field.

3.4.4 Synchronous Speed

When a three-phase current system is applied to the stator, a travelling stator magnetic field is produced that sweeps across the secondary with a synchronous speed (v_s). This will result in an induced voltage appearing at the secondary. If the rotor is short-circuited [5], it will induce a current at the secondary which will react with the stator magnetic field to produce the expected force that causes the mechanical

translation of the system [4].
For linear induction motors, the synchronous speed is defined as the speed of the travelling magnetic field created by the stator [4]:

$$v_s = 2 * \tau_p * f \tag{7}$$

where τ_p is the pole pitch (defined in the next section) and f is defined as the frequency of the stator field (Hz). The frequency of the stator field is defined as the frequency to which the stator windings are connected [5].

The design synchronous speed will be determined by the desired final speed of the motor at a given source frequency [8]. This design magnitude of the synchronous speed can be achieved with an optimised relation between the frequency of the stator field and the number of poles. The frequency of the stator field can be increased with respect to the mains frequency by using a frequency converter, which is built from power electronics. A detailed analysis of the design and performance parameters of the motor must be carried out to select an appropriate frequency converter.

3.4.5 Poles and Pole Pitch

The pole pitch is defined as the geometrical length for a single pole in a motor [8]. The pole pitch of the stator magnetic field will determine the synchronous speed of the system, v_s. Hence an optimised value must be selected to obtain the desired design synchronous speed.

Pole pitch is directly related to the length of the motor [9]:

$$\tau_p = \frac{L}{2 \cdot p} \tag{8}$$

where p is the number of poles and the L is the length of the stator. As the pole pitch increases, end effects become more severe. For pole numbers greater than ten, the speed loss due to end effects becomes very small [9].

As the pole pitch decreases and the number of poles increases, the magnetomotive force accelerating the secondary increases. However, if the pole pitch is greatly reduced, the motor becomes very inefficient [8], as the windings occupy a larger space in the stator. This implies that the magnetic field lines that pass through these windings are increased. These magnetic field lines are related to magnetic leakage since they do not circulate through the stator core. This iron core is the natural channel of circulation of the magnetic field lines and is responsible for the production of the effective magnetic field that generates the magnetomotive force.

Therefore, an optimum value for the pole pitch must be determined, as it is a parameter that influences the efficiency of the motor. When this magnitude is exceeded, the system falls back gradually into magnetic inefficiency. This optimum value can be obtained iteratively from a simulation software that indicates the magnetic field strengths and the magnetic field losses as functions of pole pitch for the given motor design parameters.

3.4.6 Slip

The traveling stator field has a velocity named sync speed (v_s) and the mechanical speed of the rotor (v_r). The stator and rotor fields are equal and opposite to each other. This creates a repulsive force between them which produces the main linear movement in the linear asynchronous motor. The rotor field will be slower than the stator due to the delay between the fields. This delay is known as the slip of the asynchronous motor. Slip is the difference in speed between the travelling stator and rotor fields. Slip can take values from 0 to 1. Where 0 there is no slip and at 1 there is full slip.

$$s = \frac{v_s - v_r}{v_s} \tag{9}$$

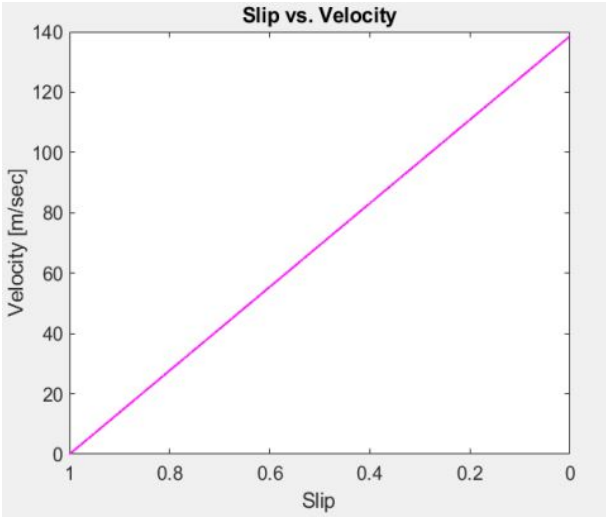


Figure 7: Slip vs v_r The sync speed is taken as 500 km/h which is equal to 138.89 m/s

When there is no slip, v_r is at the same speed as the synchronous speed. At full slip is when the largest difference between the two speeds is observed. As the rotor speed is induced by the sync speed, the rotor speed can never be greater than the sync speed. When the rotor speed reaches the sync speed it stops accelerating. Therefore, LIM is the more efficient the less slip it has. To reduce slip the phase difference has to be decreased. Phase difference decreases by decreasing the polyphase AC frequency or by increasing the electrical resistance of the circuit. Phase difference can be reduced by using a slip ring rotor instead of a squirrel cage rotor. The slip ring rotor has less slip because you can connect an end of the coils to an external resistance through slip rings.

3.4.7 Power Factor

As a project that looks to optimise its design in the future, it is vital that the efficiency of our LIM prototype is analysed throughout its development. One sound way of doing this for induction systems, like our Linear Induction Motor, is through the application of the power factor.

Power factor can be defined as the comparison of useful and wasted work done by an electrical system. It can also be seen as the phase relationship between the alternating current and voltage of a component. The relationship between the true power, which is equivalent to the useful work out of the LIM, and the apparent power, which is the total power in the system, is given by the equation:

$$\cos(\psi) = \frac{Truepower}{Apparentpower} \tag{10}$$

As this denotes to a ratio for efficiency of the electrical component, the ideal value for the power factor (PF) is 1, indicating that all of the power into the component is transferred to useful work. However, Linear Induction Motors have a reputation of having a very low Power Factor [10]; LIMs tend to have a PF of 0.8, where any power factor lower than 0.85 is seen as a poor power factor . According to Professor E.R. Laithwaite, in his paper on the improvement of power factors for LIMs specifically, the Reactive Power- the useless work done by the component- is quite high due to âthe use of smaller pole pitches and larger airgapsâ[10].

Power Factor also links the Current and Voltage, as a ratio using the phase difference between the two values. There are extreme cases in which the power factor is either 1 or 0. For example, a pure resistive load, such as a fixed resistor connected to an AC power supply, have power factors of 1, as the sinusoidal functions of current and voltage are in phase, meaning there is no phase difference. As we know that the PF is equivalent to the cosine of the phase difference, the power factor of the resistor is now 1.

On the other hand, a purely inductive load, such as a coil, have a PF of 0, since its current and voltage wave-forms have a phase difference of 0.5Ï. This can also be interpreted as the coil not outputting any

useful power, and so 100 percent of the Apparent Power is transferred to Reactive Power . The idea that the current waveform is slightly behind the voltage waveform contributes to the system being known as a âlagging power factorâ[11].

Looking more specifically at the linear induction motor, more variables must be taken into account in terms of where exactly the energy goes; the windings of the LIM, unlike the purely inductive coil, they do not have a phase difference of exactly 0.5Ï, and so possess useful true power in the form of mechanical energy transfer from electrical. This energy transfer, however, experiences some dissipation due to resistance in the coil, which can be directly associated with the resistivity of the coil [12]. Despite this, most of the power is still conserved, as the energy transferred during the maximum rate of increase of the voltage is transferred back into the system during maximum rate of decrease in the voltage . Taking into account other aspects such as the pole pitch of the motor, and the size of the layer of air between the primary and secondary segments of the LIM, we are able to use a more specific equation for the power factor:

$$PF = \cos(\tan^{-1}(\frac{\pi \cdot p_s \cdot g}{\mu_o \cdot s \cdot \tau_p \cdot \nu_s})) \tag{11}$$

Where ν_s is the synchronous motor speed, τ_p is the pole pitch of the LIM, and g is the air gap .

As a result of the power factor of LIMs being generally low, one aim for the development of this system is to improve its efficiency.

The general method to increase the power factor is by moving the phase difference ψ between the voltage and current wave-forms closer to 0. This can be done by adding in a capacitive current, which has a leading power factor [12]. This leading PF brings the lagging PF of the original inductive load forward towards a smaller phase difference. As a result, we can observe a larger power factor and higher efficiency. However, a source for this capacitive current may be an unnecessary and space consuming addition to the LIM, making it an unfeasible for our specific application.

The most appropriate strategy for this is to implement finite element analysis. This method mathematically models in detail the energetic efficiency of specific regions of the LIM, allowing analysis of where the power factor can be adjusted. In a paper written by Ch. V. N. Raja and K. Rama Sudha, a circuit and dynamic model for a linear induction motor were derived, which were then used to evaluate the power factor and optimise the LIM model accordingly [13]. Another method Raja and Sudtha adopted was the Harmony Search Algorithm. Developed to work in a similar way to improvisation, the algorithm goes through a set number of solutions, known as Harmonies, and tests them against a list of specified decision criteria. After every harmony has been applied to the decision criteria, the most ideal harmony for these criteria is isolated [14]. Applying this technique to the LIM models they had formed, Raja and Sudtha optimised the dimensions and variables of the LIM, such as p_s and separation.

Using similar techniques, the overall power factor of the LIM can be optimised, leading to a higher efficiency. This will minimise the dissipation of energy out of the system.

3.4.8 Slot and Winding distribution

The field produced by the moving stator in a linear induction motor can be ideally modelled as a sinusoidal wave made up of three phases, as it uses a three phase system. However, while producing a perfect sine wave-form would make the LIM a very efficient system without any resistive regions, the general form the three currents take is more like a step waveform, in which a sine waveform is approximated to using square waves. The horizontal regions on these square approximations of sine waves are generally due to lack of complete interactions between the windings, and so adding the number of slots in the stator, which increases the number of overall interactions the windings experience, can reduce the deviation of the square waves from the perfect sine wave . While these sine wave functions are useful, we can also look at the change in current from another perspective. The sine wave graphs that have been used are with respect to time [14]. However, with the use of Fast Fourier Transforms, a new function that instead looks at the current with respect to frequency rather than time can be generated [14]. Comparing the Fast Fourier Transforms of increasing slots and decreasing slots, we see that the form of current frequency-wise also improves, as the waveform approaches a sinusoidal wave .

Another benefit of having an increased number of slots is that there is a decrease in joule loss [15]. Joule loss refers to the dissipation of energy caused by Eddy and other circulating currents (mentioned later in the report), which is directly proportional to the square of the current in the coil and the resistance of the coil.

$$JouleLoss = I^2 \cdot R \cdot t \tag{12}$$

Also known as copper loss, for every 1 kW of energy into the coil, 1 joule of heat is dissipated from the system [16]. While this loss is not as prominent in smaller scale electronic systems, larger systems such as our LIM can experience copper loss throughout its interactions. Increasing the number of slots increases the amount of magnetic resistance, reducing the amount of overall resistive forces that cause the Joule Loss [15]. The increase in slots for a given length also increases the slot density, which means the same amount of driving force can be generated with a shorter length of stator, making the system more cost and material- efficient. Increasing the number of slots also can increase the slot fill factor [15], which describes the proportion of a slotâs area covered by the conductor, in our case the coils [17]. A larger slot fill factor contributes further to an increased power efficiency, as well as the power factor. Increasing the slot number has this effect as, if the number of slots increase on the same length of stator, the slot density increases, causing the size of the individual slots to decrease. Therefore, the coil takes a higher proportion of the area of the slot.

However, there is a limit to the number of slots that can be added as, after a certain number of slots, the flux generated in the airgap of the system becomes unfavourable, deviating the waveform again from the sinusoidal pattern [15]. These components are made up of harmonics higher than the fundamental frequency. The fundamental frequency is the harmonic that generates the driving force we are looking for. However allowing higher harmonics to also develop can disrupt the base harmonic, producing other forces that oppose the driving force. As a result, the efficiency of the motor can begin to deteriorate with high numbers of slots. A further advantage of the use of a relatively large amount of motors is that the cogging torque can be minimised by cultivating a large common multiple between the number of poles in the motor and the number of stators [15]. Cogging torque refers to the torque produced between the magnetic fields of the rotor and the stator. In motors, especially at low speeds, this can cause a huge loss in energy due to resistive forces, however, increasing the number of stators can reduce the effect of cogging torque â although this becomes less of an issue at higher speeds as the torque gets easily overcome by the driving forces of the motor [16].

Overall, we can add slots to the stator and thus allow the waveform to approach a sinusoidal form. However, exceeding a limit for the number of teeth can bring inefficiencies in the form of counteracting forces from higher harmonics that reduce the power factor by negating the driving force produced due to the fundamental frequency. Furthermore, the complexity of manufacture for a high slotting density on the stator could exceed the capabilities of the team at this point. However, through iterative testing of the stator, and running simulations via techniques such as the Harmony Search Algorithm, we can pinpoint an ideal stator slot density that will optimize the efficiency of the system.

3.4.9 Lorentz Force

Induced eddy currents in the rotor create equal and opposite magnetic fields which are attracted to the magnetic field of the stator. The two magnetic fields move at different speeds with a constant difference: the slip. The motion is linear, and the mathematical relationship describing this, the Lorentz force acting on the rotor, is given by:

$$\mathbf{F} = Q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \tag{13}$$

Where F is the force produced. Using this relationship, and the properties of magnetic fields it is found the thrust generated as a result of the Lorentz force is modelled as [8],

$$F_{thrust} = \int_0^L Re[J_z(y,t) \times B_x^*(y,t)]dy \tag{14}$$

Where $J_z(y,t)$ represents the eddy currents and B_x^* is the conjugate of the magnetic field strength.

Shown in fig. 6 is the slip speed against thrust. This demonstrates how the optimal value of a parameter can be found for our system. However, with such a large number of parameters that may have an effect on one another, a more in-depth analysis of the optimal parameters is required.

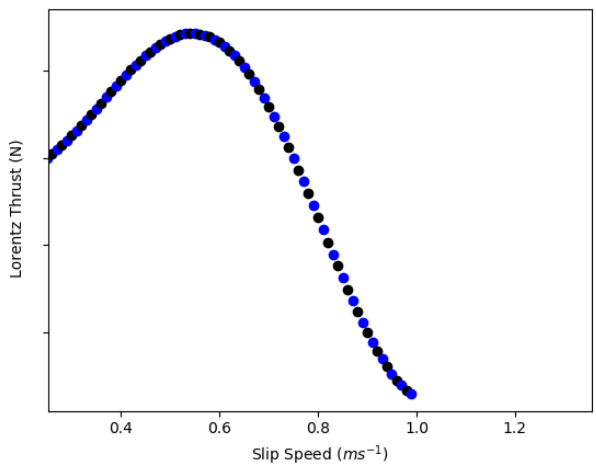


Figure 8: Slip Speed against Thrust Force

3.4.10 End Effects

It is found that as the motor is now linear instead of traditionally circular its asymmetry will result in end effects. At the open-ended gaps of the linear motor the magnetic field is interrupted, making the field of the linear motor stop abruptly. These discontinuities in the stator field heavily influence the thrust, power factor and efficiency of the motor; especially at high speeds as the magnetic field distribution is different in the air gap at the ends.

An approach to reducing end effects is to operate near the synchronous speed, as end effects can be mitigated in a low slip region[8]. Additionally, due to the thrust being relatively small in high slip regions, keeping the frequency of the motor low further optimises the system. This can be achieved by using a variable speed or frequency drive for the motor.

Altering the winding configuration by increasing the number of windings or slots per phase of the stator ensures both a symmetric and gradual phase distribution throughout the linear motor as it represents a better approximation of the lissajous distribution of the travelling field [8]. Like this, the disruption in the field is less apparent and the field distribution seems less discontinuous.

Another way of reducing end effects is to capture more of the magnetic field by implementing bigger teeth on the beginning and the end of the motor. These teeth have been seen to saturate slightly faster than the rest, which is why adding more active surface area to them helps with capturing more of the field and generates a beneficial effect on the overall thrust.

Further optimisation can be made by altering parameters e.g. by increasing the pole number (improves force capability) or keeping the air gap distance as small as possible (produces a high magnetic flux density) [1]. However, the resultant decrease in motor performance when adjusting these variables has to be taken into account and a 'trade-off' point must be found.

One method of reducing end effects is though finite element analysis and iteration. The optimal parameters that minimise end effects are found though setting an initial parameter, which becomes more accurate after each iteration. However, if the end effects are minimised it does not mean that the thrust is necessarily maximised. A similar method should be used in optimising thrust. It may then be seen that there is a 'trade off' between increasing thrust and reducing end effects.

3.5 Design and Testing

A clear design path must be set before starting any serious geometry specifications and simulations. This section will cover the proposed design process by the Propulsion sub-team. A subsection is dedicated

to the testing procedure of a prototype, it includes the variables that must be measured to determine efficiency and a proposal of a simple yet effective testing apparatus that will allow the sub-team to measure these variables.

An important consideration when designing all subsystems, but especially the propulsion subsystem, is that if a low ambient pressure is chosen for inside the tube, there will only be a small fraction of the convection cooling we would expect at atmospheric pressure. This means that thermal energy losses due to friction or electrical resistance may prove harmful to the Pod and as such must be kept at a bare minimum. To ensure a faster cooling and thus larger energy loss tolerance, outside air could be used to cool the pod's internal systems, but the effectiveness of this process will again be only a fraction of what it would be at atmospheric pressure.

3.5.1 Design process

The fundamental elements of the design process are establishing objectives and criteria, theoretical framework, analysis, construction, testing and evaluation. The specifics of prototype manufacturing will not be covered in this report. A flowchart of the design process is shown in Figure 9.

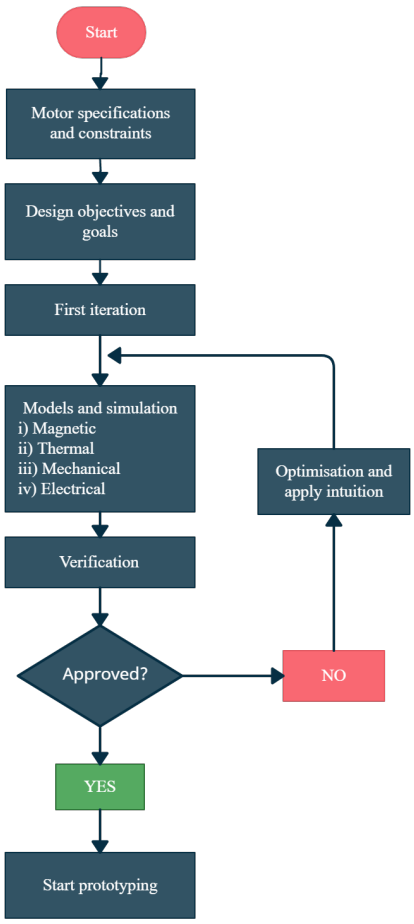


Figure 9: Design process flowchart

Motor specification

In this step key variables and design limitations must be introduced. This includes dimensional, weight and power limitations and also force requirements. These parameters tend to remain constant throughout the design and optimisation process. But since ICLoop is still at a very early stage, objectives and

requirements may change throughout the design process. The force requirement has been calculated using a kinematic analysis of the first iteration. Please note that the Propulsion system objective is to be able to provide a force that accelerates the Pod to 400 km/h in 14s, this requires a force just bellow 2000N. This force output is a key parameter for the Chassis team, which must design and build a structure that can withstand this force, amongst others.

Design objectives and system goals

These have already been mentioned previously. Once again, the propulsion sub-team objective is to develop a LIM capable of delivering a force of over 2000N, allowing the Pod to accelerate from rest to 400km/h in 14s.

First iteration

For first iteration assumptions pods previously used in the SpaceX Pod competition were researched. Rough estimates and averages of Pod parameters and trajectories were calculated in order to have a representative first iteration. From previous competitions, pods tend to weigh no more than 250 kg. It is important to note that pods using rotational motor technology will have a very different weight distribution as they must have transmission systems and wheels.

Using this initial assumption of Pod weight, together with the known length of the track and a target velocity of 400 km/h, a required force output for the LIM was obtained. This rough estimate does not take into account any friction forces, as these depend mostly on other sub-teams's future work. Acceleration and deceleration times can vary as long as the Pod has an adequate safety margin and the maximum track length is not exceeded. The weight of the Pod will be better known towards the end of the design process all the sub-teams start compiling all their data together.

Models and simulations

This section is based on the use of simulation software to computationally create and test models. Some of the simulations that will have to be carried out are electromagnetic simulations (EM simulations), thermal simulations, structural simulations and electrical simulations. First and foremost, the EM simulations will be carried out to optimise the power output of the LIM and to minimise the losses, mostly by minimising the end effects. Thermal, structural and electrical simulations will then be carried out to test the integrity of the system under the conditions it will be placed at during the competition. The thermal and electrical simulations will be closely linked, since most of the heat loss will come from electrical losses, since there will be no heat generated by moving parts such as axles or wheels. The structural simulation will be done to ensure the system can endure the vibrations and various forces it will experience in flight.

Verification and approval

This is the most important part of the design process. All the data extracted from the simulations must be processed and compared against the motor specification and constraints and make sure that the requirements are met while not exceeding any limitations.

Optimisation

Once the iteration has been verified, the model will be optimised to reduce weight and volume, as well as to increase thrust output. After this, the optimised version of the model must undergo all the testing above again, until it can be optimised no further.

Start prototyping

After the LIM model has been fully optimised the design process comes to an end and the prototyping process starts. If the prototype behaves in an unexpected manner, this may be due to some flaw in the design calculations or, most likely, due to poor manufacturing. In this case another iteration of the motor will be needed, together with another prototype. Once the prototype behaves as desired, the final version is ready.

Prototype testing

The first part of prototype testing will look at testing isolated subsystems. To test the propulsion subsystem we will use a testing apparatus consisting of a disk that rotates freely around a central axis. The setup will be completed by a set of clamps that hold our LIM prototype in place at a set air gap (distance from the disk). This setup will allow us to experimentally calculate the thrust output of the motor, as well as its durability. In order for the test to simulate the competition environment better, the test should be carried out in a depressurised chamber.

3.6 Machine Learning Optimisation

Theoretical calculations form the backbone of the optimisation process. However, the complexity of the Pod and its propulsion system requires the use of approximations in calculations. A common approximation would for example be to assume that wires are infinitely thin. While the results of calculations using such approximations are still very useful for improving our design, they will inevitably lead to a non-optimal design. The propulsion team plans to mitigate this problem through the use of numerical methods paired with machine learning.

3.6.1 Generative Design

Generative design is an iterative design process in which a design is improved through repeated trial and error [19]. Generative design is commonly paired with evolutionary algorithms and automatic testing to allow for the creation of designs with minimal human intervention. Generative design can be implemented for all tasks which allow for numerical evaluation of the design and can optimise a design based on multiple criteria. The algorithm behind evolutionary generative design is very simple and a rough pseudo code is shown below.

Algorithm 1 Evolutionary Generative Design

```
Randomly generate initial population of designs
for Number of Generations do
    Test designs
    Create new population based on test results
    Mutate new population
end for
Save best design
```

When implementing this algorithm for designing LIMs, the population will be a set of CAD-designs conforming to certain constraints on height, length, width, material composition, number of elements etc. The initial population will be created randomly without supervision.

To test the designs, the code will interface with a simulator which will determine design parameters such as weight, generated thrust and power. These values will then be fed through a set expression, to create a fitness score for each design. The expression used to calculate the fitness score will make it possible to optimise for different criteria. It could for example be $thrust/weight$, $thrust/power$, $thrust^2 - power^2$ or any other combination of the feature values.

To create a new population based on the test results we will employ operators such as elitism, crossover and tournament selection [18]. This means that a few of the top performing designs in the original population will pass directly and unchanged into the new population. Then to make the new population the same size as the old one, we will randomly select a number of pairs consisting of two designs. The probability that a design is chosen to be part of a pair is determined by their fitness score. Each pair of two designs will then be merged into one. We will do this by taking some elements of the first design and adding some of the elements of the second design. The merged design will then be a mix of the two designs in the pair. The same way the genetic make up of a child is a random selections of the genes of their parents.

To mutate the population we randomly select a subset of the population and make a random change to it. The change might consist in creating/deleting a wire between point A and B, creating/deleting a stator element at a given point or similar.

When run for enough generations the LIM designs should gradually improve and converge to a global or local maxima in design performance [19]. Due to the random nature of the generation process, the population might converge to a previously unseen design, which with luck would perform better than conventional LIM design. If the new design performs worse than conventional designs then algorithm is simply run again, until the created design has satisfactory performance. If the performance of the new design is superior to that of traditional designs we would then either try to directly implement the new design or try to find what makes the new design better and use this to improve the existing designs.

A disadvantage of generative design is that it can create designs which are very complex and hard to implement in the real world [20]. It might therefore be necessary to simplify the computer generated designs if they are to be implemented directly. The performance of the LIM design would then be negatively impacted, but hopefully it would still have superior performance to conventional designs. Moreover, even if the created designs are not realistically implementable, they might give insight into ways of improving existing designs.

3.6.2 Further optimisation opportunities

After the prototype is built, further optimisation can be achieved by leveraging Unsupervised Learning Algorithms. Unsupervised learning is a type of machine learning that looks for previously undetected patterns in a data set with no pre-existing labels and with a minimum of human supervision.

ICLoop aims to use Unsupervised learning to analyse sensory data by finding complex correlations between features including acceleration, velocity, power consumption, heat generation and unwanted mechanical vibrations. Finding these correlations enable us to discover areas of efficiency improvements and reduce wear on our propulsion system.

The optimisation can be broken down into three main stages:

Setup

During the setup stage we plan to equip both the Pod and the test-tunnel with various types of sensors. The sensors include temperature, accelerometer, gyroscope and velocity .etc.

Information Collection

During the information collection stage the Pod is ran through the loop several times while changing parameters of both the Pod and the tunnel. These can include all the variable parameters the system has for example the acceleration and velocity of the Pod The behaviour of the system (power consumption, vibration, temperature) is monitored and stored for the optimisation stage.

Optimisation

During the optimisation stage the collected data is analysed by an algorithm to find complex correlations. A potential algorithm is Simulated Annealing. It looks at the collected features, and given a required outcome (higher velocity, lower heat losses, less mechanical vibrations), it optimises it by applying a probability function in order to reach a global minimum.

This pseudo-code is a rough overview of the algorithm:

Algorithm 2 Simulated Annealing Optimisation

```
Start with a temperature T
for a finite set of iterations do
    sample new point x with certain features
    jump to a new point with probability given by an acceptance probability function
    decrease temperature T
end for
save best feature values

where
acceptance probability function = P(x, x+, T) = 1 if f(x+) >= f(x) else e^((f(x+) - f(x))/T)
```

The convergence of the algorithm to the optimised behaviour can be illustrated as shown below:

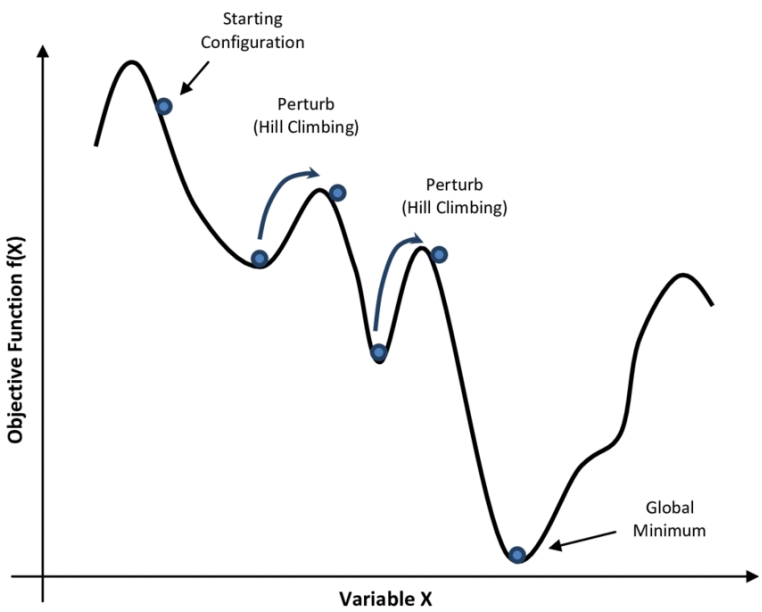


Figure 10: Convergence of Simulated Annealing where the Objective Function is the temperature.

A benefit of Simulated Annealing is that the underlying correlations do not need to be specified by a function explicitly. Instead, it finds the minimum only based on the input features and the specified behaviours to be optimised.

One challenge of this method is that it identifies correlation, however it does not present information about causation. This means, that after finding correlations we need to test a new design in a systematic way in order to find the cause of certain behaviour. An example of this can be a correlation found between lower velocities and has lower mechanical vibrations. At first glance this might appear to be due to smaller external mechanical forces acting on the Pod at those velocities. While this can be true, it might lead to a wrong assumption that our design is inherently incapable of reaching higher velocities. This is due to the fact, that the phenomenon can also be caused by electromagnetic interference of the current carrying wires/magnetic components, which, upon relocation, cease to interfere.

Therefore, our team is going to analyse the results thoroughly to ensure the best possible results.

4 Levitation

4.1 Options considered

4.1.1 Air Bearings

One of the first levitation systems that was considered was the air bearing system. The air bearing system works by storing a gas, usually air, at high pressure and then releasing it through small holes and gaps underneath the bearing to lift it off the ground, so that the bearing and any object on top of it levitates off the ground. Air bearings offer one of the least drag ratios and hence is a very good candidate for the Pod

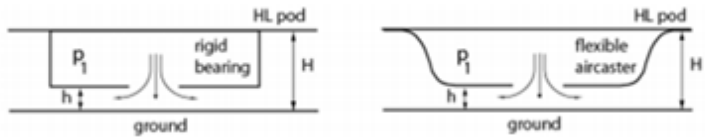


Figure 11: Air bearings can be rigid units, as shown on the left, or flexible canisters as shown on the right. [21]

The amount of propulsive power required to operate air bearings is exceptionally small compared to wheeled or magnetic systems. For a height above ground/ height of levitation (h in figure 11) of $10\mu m$, the power required is about $102W$ and about $504W$ for a h of $40\mu m$ for the bearing to operate at speeds of $100ms^{-}1$ [21]. However, there are some major drawbacks to air bearings. The values of h stated above is the maximum range in which air bearings can operate without imposing very difficult to achieve constraints. To achieve an air gap in the millimetre range, similar to the levitation heights needed for the competition, the bearings require a mass flow rate of about 150 kgs^{-1}, which is in the range of the mass flow requirement of a rocket engine. Therefore, even though promising, they were discarded as the amount of air that would be needed to be carried makes them unfeasible.

4.1.2 Wheeled systems

The traditional system of a rotational motor connected to a wheel to levitate the Pod off of the ground was also considered. This method is very reliable and simple and is known to be consistent, and therefore easy to model, cheap to design and to manufacture, and with good drag results at low speeds. A major drawback with the wheeled system is that it produces a lot of rolling friction due to contact of the wheels with track surface. This also means that it requires a lot of propulsive power for high speeds â about $103W$ for a speed of $100ms^{-1}$ [21]. Another reason why this system was not considered to be

implemented in the Pod was the fact that the hyperloop system is being designed to be scalable, and so the overall system should be able to work efficiently for any given scale. The wheeled system, although being simple and reliable, produces a lot of drag at speeds and weight of a real-life scale Pod Thus, the high speeds required would not be reached with any sensible power requirements. The wheel will also wear out and wear the track as well, which means that the track and the wheels would have to be maintained regularly. Unpowered wheels were also considered (no motors rotating them). These wheels would be used for a low-speed system when the main levitation system used does not produce enough lift, so the wheels would keep the Pod at a safe height above the track surface. The wheels would also allow easy transportation of the Pod when it is at rest i.e. before and after the test run. If used, polyurethane is considered as the material of choice for the wheels, because they have properties that make them suitable for use on a hyperloop system. These include excellent wear resistance for high speeds and long run times, a high load bearing capacity â the material deforms when under load, however, returns to its original shape once the load is removed. Polyurethane is durable even in harsh conditions like extreme temperatures between -28°C to 100°C. Polyurethane wheels are also very abrasion resistant which makes them superior wheels compared to plastic, rubber or even metal wheels

4.1.3 Magnetic Levitation Systems

Another alternative for a levitation system is electromagnetic levitation. This technology offers many advantages for a hyperloop design. This innovative concept allows the hyperloop to levitate due to magnetic forces and eliminates the friction that arises due to contact with the track. The options considered for the magnets on the hyperloop tube are linear and rotating Halbach arrays. A Halbach array is an arrangement of permanent magnets that increases the efficiency of the magnet system. Its arrangement augments the magnetic field on one side of the array while it is almost negligible on the other side. The field produced is very strong compared to other arrays.

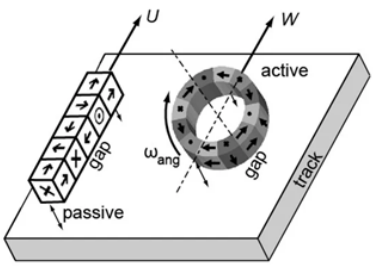


Figure 12: Linear and circular arrangement of a Halbach array.

As the hyperloop moves through the conductive tracks, two forces arise from the magnets: the magnetic drag and the Lorentz force. As the Pod moves, eddy currents are induced in the tracks. These create a counter magnetic field which opposes the change in magnetic field which cause them, according to Lenz's law. This results in a force that opposes the movement of the Halbach array, the magnetic drag. The eddy currents also cause the Lorentz force, an upwards force on the array moving over the tracks. This is the force that causes the Pod to levitate. If the magnets are strong enough, when the Hyperloop has reached a velocity high enough to produce a Lorentz force greater than the weight of the Pod, it will then levitate. This means that in the initial phase of the trajectory, before reaching that velocity, the Hyperloop would need auxiliary system of wheels, unless motion is created in a different way, such as in the Rotating Halbach Arrays.

4.2 Theory Behind Chosen Option - Rotating Hallback Arrays

The rotating Halbach array can be considered as a derivative of the linear passive maglev. The two work and produce lift and drag in a similar way. The rotating Halbach array is a circular version of the linear Halbach array that uses rotatory motion to generate lift. It does not need to be moving along the tracks, it can levitate while stationary in the conductive tracks. Using a rotating Halbach array therefore

eliminates the need for the auxiliary system of wheels but needs a high torque motor. These are more expensive and complex and has a relatively high energy consumption.

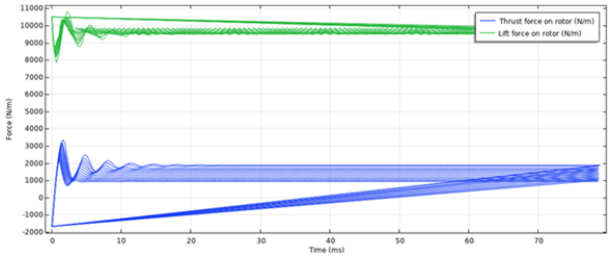


Figure 13: The lift and drag of the rotating Halbach shown in velocity increments from 10 ms^{-1} up to 100 ms^{-1}.[23]

Halbach arrays are a special arrangement of magnets that results in a stronger magnetic field on one side and the magnetic fields on the other side end up cancelling out. If we were to arrange a Halbach array into a circular geometry and connect it to a motor, we could create a rotating Halbach array which could be used effectively to produce lift. The drawback of a Halbach array however is that it is quite complicated to manufacture and assemble and produces a lot of heat in the metal that it rotates above (the metal that its magnetic field interacts with). This is due to the drag produced by the movement of the eddy currents that the magnetic field generates. If these are mounted on gimbaling mechanisms, then they can be positioned at an angle to produce thrust as well, moving the Pod forward.

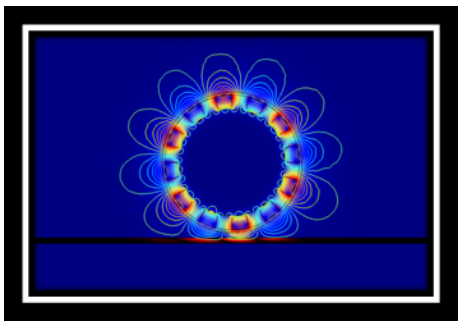


Figure 14: The lift and drag of the rotating Halbach shown in velocity increments from 10 ms^{-1} up to 100 ms^{-1}.

As can be seen from the above graph, a rotating Halbach array can produce about 10000 N of lift force, which is more than enough to lift up the Pod at the scale being considered. The Halbach arrays used to produce the graph used 24 Neodymium magnets which massed over an aluminium track. The Halbach array, however, was tested with its width facing the edge, rather than its face, as shown in figure 3. The numbers however are still promising and show the potential of the rotating Halbach array as an excellent choice for a levitating system, and perhaps even a propulsion system, if designed in such a way.

More simulations and analysis will be required using software like Ansys and Comsol to produce data that can show the performance characteristics of the rotating Halbach array. This will also confirm the advantages of using the Halbach array as a levitation method for the hyperloop Pod and give an indication of how the array needs to be designed for optimal performance.

5 Suspension

5.1 Introduction

The first thing that must be considered are the requirements for suspension that a vehicle of this characteristics would have. What kind of suspension would it need, if it needed any? If there was unevenness in the track, this would momentarily decrease the distance between the levitation system and the track. This would lead to an increase in the lift produced by said system, and in turn increase the distance between track and system past the equilibrium point. This shows that just one point of unevenness would lead to oscillations in the vehicle, like those that would be caused by a spring, which would have a knock-on effect on the aerodynamics, propulsion system, and in general the performance of the Pod This makes the inclusion of this system a must. Additionally, because the Pod is not being held against the track in any way, there must also be a system which deals with sideways oscillations, as otherwise the Pod could lose it's direction and move away from the track, which would be catastrophic as the Pod hit the wall.

5.2 Car Suspension

Due to the similarities in frame structure between a road vehicle and an Hyperloop vehicle, which will be explored more in depth in the modelling subsection, the IC loop team researched on road vehicle suspension designs as reference for their design.

Car suspension designs can be classified into three types: beam axel, Macpherson strut, and double wishbone. Among those designs, Beam axle is known for being structurally rigid and is mainly used on off road or heavily loaded vehicles; Macpherson strut is known for its low manufacture cost, and the double wishbone structure is known for its performance in absorbing bumpiness from uneven surfaces on the road and is widely used in Luxury cars. [35]

Beam axle suspension consists in suspension attached to a beam axle that ties the two front wheels. The connection to the chassis is made by link rods that allow for vertical movement of the suspension and also serve for longitudinal suspension, and a single lateral locating rod for each beam (2 total) that stops lateral movement but allows for vertical movement. There are 2 springs, one on each side of the axle, as close to the wheels/skis as possible which serve as shock absorbers, and dampers can also be introduced. In general, they are simple and generally low construction costs. They are very good at providing high strength but imply extra weight (multiple large rods included). The way this is connected to chassis has a big impact too, possibly increasing the loading on the screws .

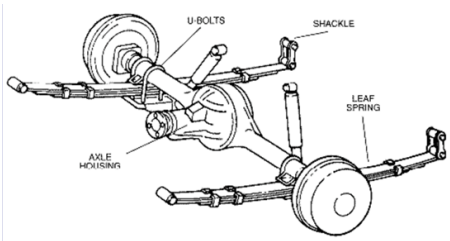


Figure 15: Example of a leaf suspension, based on a beam axle suspension. [34]

Meanwhile, the MacPherson Strut Suspension use lower A-arm and spring/shock absorbers to connect wheels/skis to the chassis. Both provide lateral and longitudinal locating while allowing vertical movement. They are low cost and relatively easy to construct. This requires more vertical height for the suspension packaging due to the strut and spring layout, which might not work for our chosen design.

As the suspension of hyperloop vehicle would only experience maximum of 1mm displacement on track, and aims to maintain steadiness of the cabin, the double wishbone structure should be chosen as a starting point.

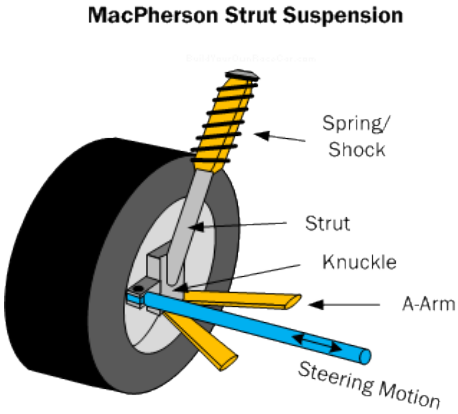


Figure 16: Example of a MacPherson Strut Suspension.[28]

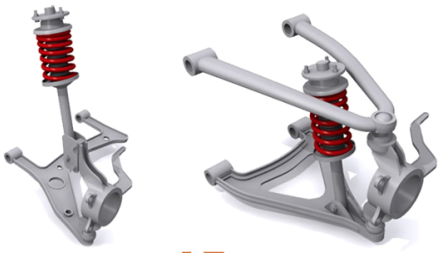


Figure 17: Example of a wishbone suspension.[34]

The double wishbone suspension (shown in figure 17) holds the axle on upper and lower ends, therefore enabling the axle to stay horizontal to the ground during vertical displacement. This consistency means that the structure would be able to easily hold the magnet horizontal to the ground, which reduces possible out-of-plane oscillations due to part of the lift force not being perpendicular to the rail.

Even though car suspension serves as a good starting point, the structures should be simplified before being used in a hyperloop, as hyperloop vehicles need minimal steering compared to road vehicles, and have a large weight constraint.

5.3 Modelling the Suspension

Firstly, some key factors must be defined for the Hyperloop like an approximation of the final mass, as well as the design suspension gap height. Knowing these parameters, the next step would be to approximate the system and the force that would act on it. A relatively simple 2 mass system approximation with 2 springs - one corresponding to the actual suspensions and dampers, and one corresponding to the electrodynamic suspension which acts similarly to a spring, can be used. Knowing this, the system's dynamic response equation can be written. Simultaneously, the force applied on the system by the track can be approximated, this will be the force that we will need to design against. From the competition specifications, teams are told steps between track sections (every $12.5ft$) are up to 1 mm in height. However, as can be seen from the MIT report [33], the track is rougher than defined, and it might be useful to design for slightly larger bumps. The reason an idea of the gap height is needed prior to this, is because the suspension system should allow the skis to maintain the ideal gap height as it crosses over such disturbances. To avoid failure the vibrations introduced to the system must be smaller than the gap height, since collision between the skis and the track must be avoided. With these factors in mind, the strength of the disturbance in the system and the subsequent amplitude of the wave can be approximated for a range of stiffness and damping values. As such, these values can be plotted to find the ideal behaviour.

After that has been completed, an initial design for the springs and struts should be made, based on the

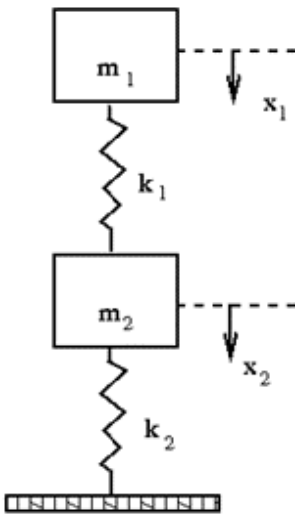


Figure 18: 2 mass spring system approximation –Maglev suspension assumed to work similarly to a spring –Mass 1 corresponds to the Pod, Mass 2 to the skis –A damper of value c1 would be added between Mass 1 and Mass 2 [32]

research carried out into car suspensions or other suspension systems [30]. The influence of geometry, including the optimum angles at which the springs and dampers operate, and their influence on the spring stiffness ratios, must be looked at.

The various directions of motion must also be considered when designing said suspension. There is a need for both pitch and heave control which would account for vertical and longitudinal movement (pitch corresponds to movement of the Pod as it decelerates and the force is transferred towards the front, and heave to the acceleration movement, as the Pod speeds up) and lateral control. It should be noted, however, that lateral suspension needs to be treated separately as it cannot be modelled using the same 2 mass system and its movement is mostly, if not completely, independent of the vertical and longitudinal movements.

The Lateral Suspension‘s design process would be very similar to the pitch and heave suspension. The main difference would be the fact the suspension will rely on the I-beam placed in the middle of the track as a way of centering the heavily prone to lateral motion levitating Pod This will require an interaction with the I-beam which will be likely to cause some form of drag (whether due to wheel friction, or magnetic drag from magnets). The possibilities could be to use a regular suspension system with wheels, like a car, or other methods, like electromagnets, skids, or permanent magnets. Skids are likely to be insufficient and would cause too much friction at high speeds. They are also problematic since they will require frequent changes and need to be made of a material softer than the track. On the other hand, electromagnets are precise but would require an additional motor and input, which means we would need to differ from our initial approach using permanent magnets and electrodynamics suspension. Thus, the two options left are wheels and permanent magnet suspension. Wheels are easier to design but might be more costly. Their performance will be good at all speeds, however. On the other hand, permanent magnets are efficient only at high speeds, which means a low-speed mechanism is recommended. Additionally, they would involve additional design time.

Once the geometry has been chosen, digital modelling and simulations can be started. Using computer software like SolidWorks, a 3D model of the chosen design can be rendered and proceed to be tested in structures simulation software using Finite Element Analysis (FEA) methods. To do so the skis should be designed and added, for optimum results. The mass of the Pod could simply be added as a vertical force on the skis to simulate the entire Pod Then, the force in the worst loading scenario should be applied, to estimate the worst case loading on the system. These simulations should also measure torsional loads. It will be observed how the system behaves at the front and the back of the skis (most acute pitch and heave movements) as well as its global vertical acceleration and displacement oscillations. For the lateral suspension, the springs and dampers should be tested for lateral and rotation motion exclusively,

Figure 19: SpaceX Hyperloop Competition I-beam track layout with some irregularities [31]

but that is the only difference. The values collected should be the same. The response of the system at various speeds for varying values of springs and dampers should be observed, to obtain the best frequency response of the system for corresponding suspension values. If the amplitude of the oscillations for the system is lower than the ideal gap height, we will have met the requirement that the skis shouldn't touch the track at all points. It can be deemed reasonable that the suspension system will hold for the ideal values of spring stiffness and damping that we will have found from the iterative testing processes.

6 Braking

6.1 Options Considered

There are several different types of braking systems that were considered for the hyperloop design. The possible types of braking systems that could be implemented were eddy current braking, friction braking, and braking using a linear induction motor or linear synchronous motor.

Eddy current braking involves using eddy currents that are created when a conductor passes through a magnetic field. The eddy currents create forces that oppose the motion of the conductor. Thus, the opposing force can be used as a braking force. The braking force is a function of three main parameters –relative velocity, distance from the magnet and the configuration of the magnetic array.

Figure 20: Braking force vs velocity characteristics (F_L is lift, F_D is drag) [36]

The graph above shows that at low velocity, the drag force is greater than the lift force but as velocity increases, the lift force becomes greater than the drag force. Because the braking force is directly proportional to speed, and since at high speeds the lift force is greater than the drag force, an additional set of wheels might be needed to stop the vehicle.

Friction braking involves the conversion of kinetic energy to thermal energy by applying friction to the moving parts of a system. In a hyperloop, brake pads would come in contact with the I-beam, which would cause the hyperloop Pod to slow down due to friction. The brake pads could be clamped onto the I-beam during braking by using either spring force or linear actuators. The brake pads would need to have the maximum possible surface area, to create the greatest possible frictional force to slow down

the Pod quickly. Furthermore, the material of the brake pads would need to be long-lasting and heat tolerating to ensure the brakes do not get damaged easily but would also need to be softer than the I-beam.

Friction braking has several disadvantages, for example, there is a risk of damaging the I-beam due to the frictional force and it is also very difficult to simulate the interaction between the I-beam and the brake pads at extremely high speeds, from which the Pod would be slowing down. [33] Therefore, friction braking would not be the best option for the braking system in the hyperloop Pod

A linear induction motor could also be used as the braking system. A linear induction motor would usually be used to accelerate a vehicle, however, if the polarity of the electromagnets in the stator is reversed, deceleration can be produced. A LIM consists of a primary and secondary part. When power is supplied to the primary, magnetic flux is created and induces eddy currents in the conducting material of the secondary. The magnetic flux in the primary and eddy currents in the secondary interact to produce a linear force, which can be used for deceleration. [37] The advantage of using a LIM for braking is that no gears are required, and that high accelerations and decelerations can be produced. There are two types of LIM that could be implemented in a hyperloop, short primary and long primary. In the long primary variant, the guideway is the primary and the secondary would be on board of the Pod In the short primary variant, the opposite occurs and the guideway acts as the secondary while the primary is on board of the Pod If the short primary is used, then on-board power is needed to generate the varying magnetic field, which will increase the cost of an individual Pod but would decrease the cost of the guideway as no active parts are required. If the long primary is used, the cost of an individual Pod decreases significantly because there are no active parts that are required, but only conductors that act as the secondary. Therefore, the long primary would be more efficient and could reach speeds of 100 km/h efficiently unlike short primary. However, using a LIM has several disadvantages, for example, it has poor efficiency and a low power factor.

A linear synchronous motor is a type of electric motor in which mechanical motion is synchronous with the magnetic field. An LSM would consist of coils which would induce a magnetic field that travels along the track and permanent magnets on board of the Pod LSM is more energy-efficient than LIM but it would be very difficult to implement an LSM because it would be very difficult to work out the exact positions and velocities of the pods in the hyperloop at any given time.

6.2 Theory behind chosen option

Passive electromagnetic brakes can be used to slow down the Hyperloop capsule from a high speed and bring it to a stop. According to Lenzâs Law, an eddy current produces a magnetic field that is in opposition to the magnetic field that produced it [26].

While moving along the rail, permanent magnets cause eddy currents. The eddy currents create their own magnetic field, which counteracts the magnetic field of the permanent magnets. As a result, a perpendicular force repels the permanent magnets from the aluminium rail. Parallel forces also counteract the movement of the magnets, causing the Hyperloop Pod to slow down. This braking system enables the Hyperloop Pod to stop without requiring contact with the I-beam, therefore reducing concerns of damage to the beam or the Pod itself [27].

The Pod uses eddy-current brakes to come to a safe stop. The brakes stop the Pod by moving a large array of permanent magnets into close proximity with the central I-beam in the track. The relative motion between the magnets and the I-beam induces eddy-currents in the I-beam, which produce a magnetic field which retards the motion of the Pod Eddy current brakes allow us to bring the Pod to a stop without requiring contact with the I-beam, thereby allaying concerns of damage to the beam or the Pod due to high contact forces.

The maximum stopping rate of the Pod should be 2.4 G of braking force with a 0.5 second engagement time. The mass of the brakes should be about 20% of the weight of the Pod

Figure 21 shows that the brakes can be divided into three subsystems: The magnet arrays, the motion control system and the suspension. The function of the magnet arrays is to generate eddy currents while

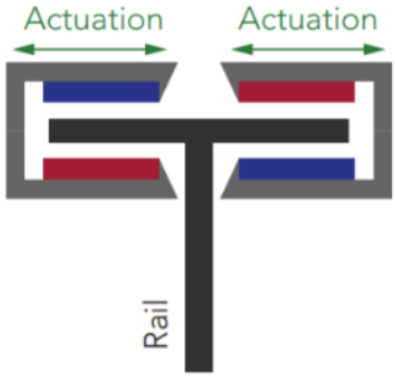


Figure 21: A figure showing the rail of the hyperloop

the motion control system moves the magnet array relative to the beam and the suspension isolates the brakes from the Pod The eddy current brakes have a lower risk of damaging the I-beam because they do not come into contact with each other, thus making the brakes very low-maintenance.

The advantages of using eddy-current brakes are that they are quiet, wear-free and friction-less. These factors make them much more attractive than friction brakes which need to be inspected regularly and friction brakes can wear out gradually [33].

To calculate the number of magnets needed for a specific force, the wavelength is to be minimised for minimum drag. These magnets are therefore small resulting in a very short wavelength in this design, and over 200 magnets are needed in the entire braking system.

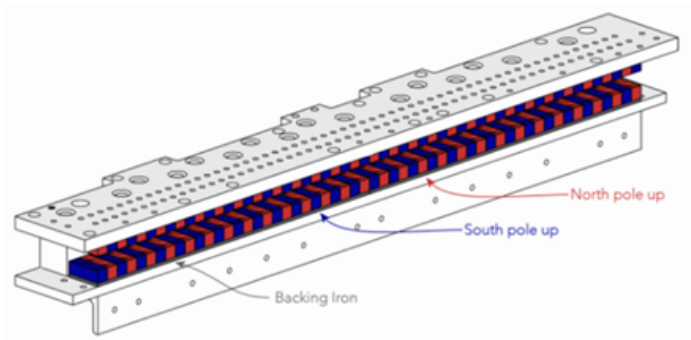


Figure 22: A figure showing the array.

The redundancies that we need to put in place in case of failure is that we must be able to modulate the braking force so that we can stop within a 50 feet section at the end of the track.

7 Electronics and Software

The Electronics and Software sub-team comprises of 10 active members across various engineering disciplines including Electrical and Electronic Engineering, Aeronautics and Computer Science. The team is in charge of delivering all power, control and telemetry requirements on the Pod, critical to its successful operation. Through the course of this project, the team will make use of software including KiCAD and Fusion 360, and languages including Python (with suitable frameworks) and embedded C++.

This chapter outlines the design choices made by the team based on the mission criteria, an overview of the system architecture, details on the navigational system, and the expected timeline. All source code is available at the ICLoop GitHub repository under an open-source MIT License.

7.1 Design Requirements

The latest SpaceX Pod design requirements available to the team are from 2018 due to restricted access; the team has designed a flexible system assuming these as an initial target and will refine the system based on the 2021 requirements as they are made available. Much emphasis is placed on Pod safety; the Hyperloop tunnel and infrastructure at Hawthorne is expensive, and ICLoop's priority is to ensure the Pod in flight does not inflict or take damage. The system needs to be designed for operation in a vacuum environment at the worst-case states (loads, temperatures and duration). Critically, the battery management system needs to be designed to tolerate these conditions, fault-tolerant in charging and can isolate the battery in off-nominal conditions. The system will also need to incorporate redundancies in critical areas, including control and telemetry.

As of 2019, all Pods must include their own telemetry hardware, retiring the SpaceX Network Access Panel (NAP). Whilst no confirmation yet, the team is assuming the tunnel WiFi will remain in place. The Pod will a strict network access protocol enforced by SpaceX including limits on bandwidth, latency and frequency range. The private network IPs are also constrained, with a static 8-bit subnet provided (192.168.0.0/24), of which only a range are cleared for use. The network should also not be bridged to the Internet in any configuration for security reasons, and the network must not run any servers or services.

To aid navigation in the tunnel, reflective circumferential stripes are lined along the tunnel walls at defined intervals. The Pod should keep track of the absolute position by counting these. When the deceleration zone is reached, the pattern change should be detected and a state change commanded. The Pod, at a minimum, is required to transmit the position relative to the start, velocity, acceleration and battery voltages, currents and temperatures. The Pod state should also be reported to SpaceX computers in a single telemetry frame (to 192.168.0.1:3000).

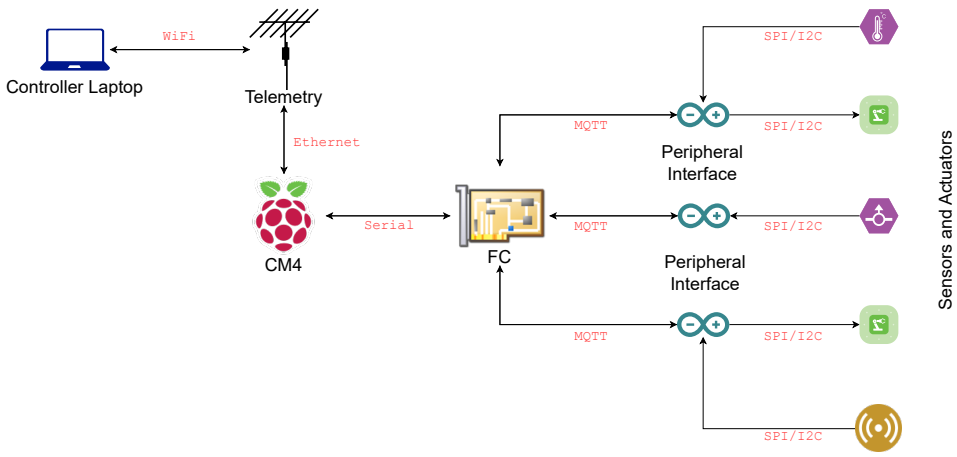


Figure 23: Conceptual architecture of control on the Pod

7.2 Proposed System Architecture

At the heart of the system is the Raspberry Pi CM4 Flight Computer - its function is to track the mission state of the Pod (idle, fault, ready, pushing, coast, braking) and the health of every subsystem, and will also manage the telemetry feedback to the controller laptop and SpaceX. The CM4 is mounted on a custom PCB, the Primary Flight Controller (FC), which is in charge of calculating the Pod state in real time and commanding the actuators autonomously. The Pod will feature an Ethernet 'star topology' network connecting the CM4 and FC to all peripheral boards. The flow of information is as described below (see figure 23):

1. Raw data is generated at the sensors connected to the peripheral interface boards
2. Each peripheral board's MCU handles filtering of the data (pre-processing)
3. This data is pushed to an MQTT broker over Ethernet, and is pulled by the FC using MQTT
4. With this, the FC calculates the current state of the Pod and relays the information to the CM4
5. The mission state for the next instant is computed by the CM4 and informs the FC
6. The FC commands the actuators as per the received flight state

In the case of an off-nominal reading from any subsystem, the FC has the authority to initiate an abort. Aborts can be commanded manually from the controller laptop, or autonomously from the CM4 or FC, and this is in line with the safety recommendations from SpaceX.

7.3 CM4 Flight Computer

The Raspberry Pi Compute Module 4 is a system-on-module variant of the widely popular Raspberry Pi 4. It has a very minimal form factor and was especially designed for deeply embedded systems, exposing all the peripheral connections through two 100-pin mezzanine connectors. Featuring a Broadcom BCM2711 chip (Quad-core ARM Cortex-A72 processor), it can run Linux-based operating systems and thus can serve as a powerful flight computer to run the Finite State Machine (FSM) and Telemetry. It also supports H.264 video encoding at 1080p30, and also very conveniently exposes a single lane PCIe port for accessing high-speed peripherals such as NVMe memory, USB 3.0, etc.

7.4 Primary Flight Controller

For the CM4 to evaluate the mission state, it needs an accurate measure of the current state of the Pod; this includes variables like Velocity in each axis, heading, temperatures, power consumption, etc. These are all calculated by the Primary Flight Controller (FC) in real time from sensor data coming from throughout the Pod. The FC needs to accommodate the power requirements of the CM4 and break out its peripheral connectors such as USB2, ethernet, PCIe, et cetera, but will also need to host a powerful MCU to perform its tasks. The ESP32 and STM32 chips are the main candidates; both can support the full TCP/IP stack and can connect to the Pod LAN, they are compatible with the Arduino codebase for easier programming and both support libraries for maths operations and for many common digital sensors.

Two prototype boards with minimal capabilities will be developed by the team in early 2021 to test the capabilities of each MCU to help make an informed choice on which to use. The interface to the CM4 will also be tested, after which the prototype boards will become handy in testing the FSM and telemetry software. The boards will be derived from the Raspberry Pi CM4 IO board, essentially a breakout board for the CM4, the source files for which are freely available in KiCAD format.

7.5 Peripheral Interface Board

The Pod will have in total around 30 sensors and actuators measuring various quantities throughout the vehicle and driving various systems. Each sensor value needs to be filtered and corrected before it can be used by the FC for accurate state approximation. For the scale of the Pod being designed, it is therefore not a sensible approach to connect every sensor and actuator directly to the FC - it would be too much load on a single MCU. The peripherals are instead wired to various boards distributed across the Pod, the Peripheral Interface Boards, each with an MCU tasked with pre-processing the data and packaging it into organised frames, in addition to controlling the actuators connected based on feedback from the FC. These boards would send the processed data to the FC, reducing the load on the latter. This would also remove the dependency that the FC would need to use API libraries to interface with each sensor. As of writing, the exact sensors have not been selected as many are dependant on the Propulsion, Levitation and Braking systems which have not yet been fully designed. Therefore, these boards will initially need to be designed to accommodate any combination of generic sensors, analogue or digital.

A prototype board will be developed in early 2021 which will include an STM32 controller, power regulation to 3V3 and 12V, an Analogue to Digital Converter (ADC) with 8 channels and 15 bit resolution, two sample sensor ports for both analogue and digital, as well as an Ethernet port. This board will be used primarily for testing the circuitry and sensor processing code, after which the full-feature board can be developed.

7.6 Navigation

The Pod will be travelling along an evacuated tunnel with well-defined geometry, for which an accurate model describing its motion can be developed. It is, however, important to have an accurate real-time awareness of the state of the Pod, especially so any abort condition can be identified as soon as possible. The navigational algorithms will run on the FC and is essentially an Attitude and Heading Reference System (AHRS) and will include sensor fusion (with Mahony, Madgwick or complementary filters, or possibly into a trained neural network) from Inertial Measurement Units (IMUs) and 1D accelerometers. The FC will keep a count of the stripe count for absolute positioning, as well as monitor the battery and internal temperatures. The data frames are sent to the CM4 and are also logged in flash memory for post-analysis.

7.7 Finite State Machine

There are only a certain sequence of states the Pod can exist in during its flight. These include idle, fault, ready, pushing, coast and braking, as well as other pre-flight test states. The Pod can transition to different states based on certain events. The CM4 is in charge of tracking the state and will do so through a Finite State Machine (FSM). It will be implemented in Python alongside the Pod health check and telemetry programs. The team will commence development in early 2021.

7.8 Power Management

The electronics, propulsion, levitation and braking systems will all require power from a battery source located on the Pod. Due to much uncertainty on the power requirement of the propulsion system, the batteries have proven to be the hardest to size accurately. However, the team has looked into the chemistry of various battery and comparing them with regards to peak discharge characteristics, heating in a near vacuum, procurement, cost, etc. The team will design and run tests on two shortlisted batteries; Lithium Polymer (LiPo) and Lithium Ion (Li-ion) in a vacuum chamber to further understand the limitations of both types and to make a more informed choice. The battery management system is entering its design phase; it is responsible for the safe handling of the batteries at all times, providing accurate and real-time information on their health and is capable of isolating them in the case of an off-nominal situation.

Additionally, the team is looking into suitable cable solutions for efficient power transmission across the Pod. Currently, the longest cable required is estimated at 5 meters; tests will be designed to decide which cable type provides the best overall compromise of least power loss, can sustain the peak temperatures at peak discharge and are as light as possible.

8 Aerodynamics

Aerodynamics governs the airflow around the Hyperloop Pod, and thus is key to the development of the Pod. Our initial approach focused on identifying key limiting parameters, which we could then minimise the impact of through numerical optimisation and computational fluid dynamics (CFD) simulations.

8.1 Background Theory

The two main components of aerodynamic drag at subsonic speeds are skin friction drag and pressure drag. The ratio of skin friction drag to pressure drag at high flow speeds is relatively low, owing to the phenomenon of flow separation [33]: this is where the airflow initially follows and adheres to the shape of the Pod, creating a thin "boundary layer", but eventually separates from the Pod. Within this layer, owing to the effects of viscosity, the flow is slowed to a stop as it approaches the surface of the Pod. This conflicts with freestream pressure, which is driving the flow past the Pod. At a certain point, the viscous shear forces opposing the freestream flow are not strong enough to retain the flow on the surface of the Pod, causing the boundary layer to no longer adhere to the Pod [33]. The ratio of inertial (freestream pressure) to viscous forces can be calculated using a dimensionless parameter called the Reynolds number (Re), defined as:

$$Re = \frac{\rho U D}{\mu}$$

where D is the characteristic length, U is the freestream velocity, ρ is the fluid density and μ is the dynamic viscosity. Re is also used to characterise whether the flow is laminar or turbulent. Flow separation causes a large turbulent wake to be produced aft of the separation point, which produces a large increase in pressure drag [33].

Using a reference Pod length of 1 m and 100 m/s as the freestream flow speed, Re was found to be on the order of 10^5. At this Re and without any disturbances on the surface of the Pod, the boundary layer would be laminar [33]. Fortunately, flow separation can be delayed by "tripping" the boundary layer such that it becomes turbulent through increasing surface roughness or creating irregular features on the surface of the Pod [54]. This is because turbulent boundary layers are better at handling adverse pressure gradients and hence can adhere to the Pod surface for longer. This delay to flow separation also means that the resulting turbulent wake is smaller than would be otherwise, reducing the pressure drag spike as noted previously [33]. As such, the team's design philosophy was to focus on a design that could maximise the delay to separation.

An additional parameter for consideration was the Kantrowitz limit - this is a thermodynamic limit regarding choked flow arising from the Venturi effect, where the rate of mass flow behind the Pod would stay constant regardless of the pressure in front of the Pod if the flow between the Pod and the tube reached Mach (M) 1 [33]. If this limit was reached or exceeded, the drag on the Pod would increase substantially.
The easiest method to prevent the Kantrowitz limit being reached is to reduce the blockage ratio (defined as the ratio of Pod frontal area to tube cross-sectional area). Since the team's target velocity of 100 m/s was approximately M = 0.3, as well as the estimated blockage ratio being far below 1, it was therefore decided that there was no need to consider this limit, as it would not be exceeded within our scope of work.

However, a target Mach number of $M = 0.3$ does present another issue - the effects of compressibility. The theoretical limit from literature for where flow can be considered to be incompressible is $M < 0.3$ [53]. As the target speed of our Pod just sits on this limit, it was decided that the flow could be initially assumed to be incompressible, with the effects of compressibility taken into account through Prandtl-Glauert corrections that can be taken to be valid through $M < 0.7$[56]. This could substantially reduce computation time for any CFD simulations, as far simpler equations can be used to solve for incompressible flow as opposed to compressible.

8.2 Methodology

Initially, the team decided to avoid parametrisation and similar methods such as using shape functions, as it was deemed that this would unnecessarily limit the Pod to a specific class of shapes. Thus, the team launched an attempt to use CFD as the main design driver: the initial concept drawings from the team were realised using the Solidworks CAD (Computer-Aided Design) software, with the goal of iterating on these designs through comparisons of CFD results. It was decided that Star-CCM+ would be used for the CFD analysis, as it was an industry-standard tool with necessary adaptability and accuracy.

Unfortunately, we discovered two main issues with our methodology: the first being that the CFD setup was time-intensive and hence not conducive to rapid iteration. This meant that, although we were able to familiarise ourselves with the operation of Star-CCM+ and select reasonable parameters for simulation (see Appendix C), it was deemed infeasible to use CFD for every iteration of our design. The decision to start with 2D simulations, although already a large simplification, did not significantly change the feasibility of frequent CFD use. The second problem was that our design space was simply too large - with no analytical method to design the Pod, there was no structure to our iterations, and hence this was an inefficient design methodology.

As such, a team decision was made to pivot to the use of parameters, which we will focus on numerically optimising before coming back to CFD for more detailed analyses. Our first iteration of parameters is shown below:

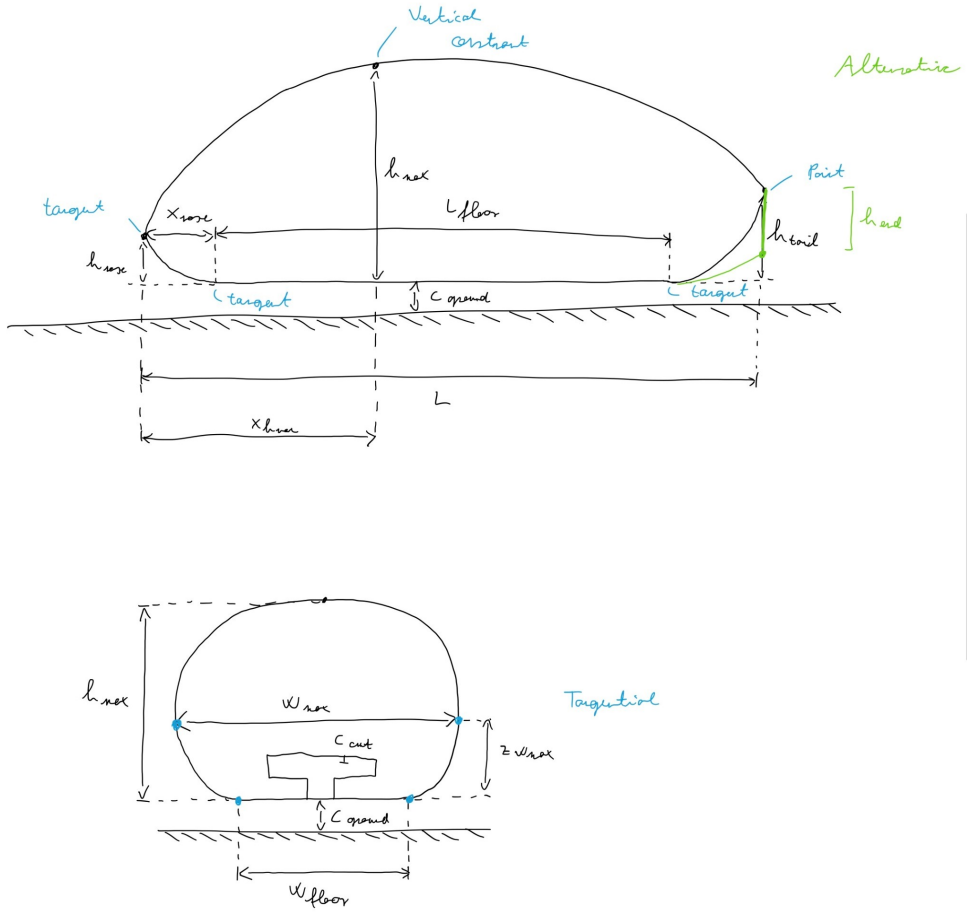


Figure 24: Parameters v1

9 Chassis and Integration

The Hyperloop Chassis and Integration team is responsible for the design and manufacture of the mainframe of the Hyperloop system; it is the base on which all the elements are supported, forming the whole Pod.

The main challenge is to design the frame for the best stiffness and strength for the lowest weight possible, as well as cost and manufacturing complexity. The structure must be capable of withstanding all the static and dynamic loads to which it will be subjected (mainly in the axial direction), as well as interfacing with different elements of the Pod which will be integrated to it. Therefore, the design considerations taken into account for this are: minimising the weight and cost, maximising the strength and stiffness, minimising manufacturing complexity, flexibility during texting for any possible change and safety to handle.

Integration is key for any engineering project. Considerations for this part of the sub-team have not been considered for the moment since at this stage of the project the objective was mostly research.

9.1 Materials

There are many factors to consider when choosing a suitable material for the chassis. Carbon Fibre Reinforced Polymer (CFRP) is a material of high preference particularly for its high strength and stiffness to weight ratio, the material tends to be stacked in a laminated form to achieve different levels of mechanical properties in the desired directions to eliminate as much weight as possible whilst still maintaining the rigidity and structure for certain applications. CFRP consist of many bundles of carbon fibres twisted into 'tows' and aligned in a particular direction to cure in resin (typically epoxy) to form plies. Few limitations include cost/availability, difficulties with additional modifications (such as drilling and bolting in extra components) and can be labour intensive. As a result, other materials were also considered during the material selection process that included a form of aluminium alloy. More advanced materials including Arovex, a variation of Carbon Fibre (CF) reinforced with carbon nanotubes graphene, and GLARE (aluminium and glass fibre bonded).

As there are many variations of CFRP, the following parameters make up the choice for the suitable material:

Cost

Carbon fibre is generally very expensive in comparison to it's competitors that offer lower mechanical properties for a substantially better rate. The material availability and labour intensity are the primary contributions to economical challenges. As previously mentioned, bundles of fibres can range between 1k-12k filaments in a tow where the 'k' represents 1000 filaments and usually lower tow content are favoured for their high strength low density properties. Although, 1k and 2k tows are not as common due to difficulties with precise manufacturing methods and tend to be primarily used for high expense applications. Many industries tend to opt for 1k-6k tow content, while the most common and widely used is 3k for the automotive and aerospace sectors as it has the highest abundance of the rest and provides outstanding properties.

Preparation methods

There are 3 main ways to fabricating CFRP; Resin infusion, Layup and Dry prepreg, each having their advantages. Resin infusion is a technique that makes use of CF fabric that are laid up into the desired configuration and by injecting resin under a vacuum to evenly distribute the viscous liquid throughout

the cross section. This method tends require additional equipment and usually a process for batch production. As only one model will be made, this leaves layup and prepreg. [38]The layup method consists of stacking dry fabric onto a mould of desired shape, similar to resin infusion, however, the resin is manually spread across the fibres instead to generate the final composite. As a result, this can introduce regions of low and high resin content and uneven, thick layers that can induce additional weight or stress concentrated areas. [39]The ideal method would most likely be the use of prepregs as the mould will be relatively easy to produce and it is essentially using partially cured sheets of lamina without the need of epoxy/resin as it has already been infused into every sheet. The only requirement is moulding the shape and leaving it in an autoclave at specified conditions for a set time to cure.[40]
Below are examples of Resin infusion, Layup and Dry prepeg:

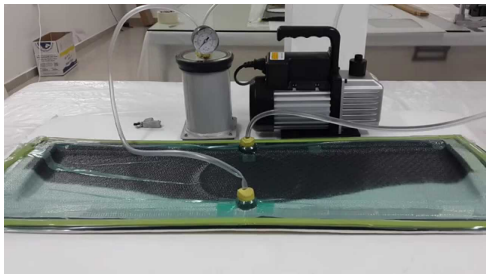


Figure 25: Resin infusion
[41]

Figure 26: Layup method
[38]

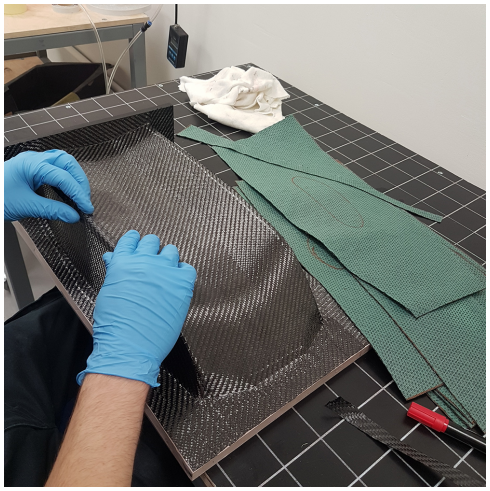


Figure 27: Dry prepeg method
[42]

Long and Short Fibres

It was decided that long, continuous carbon fibres were used in ply form where these can be laid up into a laminate accordingly. Short Fibre Composites (SFC) are generally not of much interest due to being inferior in mechanical properties in comparison, this will not be able to provide the sufficient strength at low density. Although, it is still worth considering SFC as an alternative, here is why; the handling and processing of SFC are renown to be much easier to accomplish which can be beneficial for use in this particular prototype, SFC is a cheaper option compared to continuous fibres and finally, SFC can provide strength in all directions by random fibre arrangement in one single ply, this eliminates the need for multiple layers of material (however, the parts are usually very thick and dense). The figure below highlights the geometry of continuous fibres on the left and SFC on the right with random fibre arrangement through its entire structure. Particles can also act as a form of reinforcement for composite material, these are less often used but can be quite effective especially in comparison to a bulk, non-reinforced material.

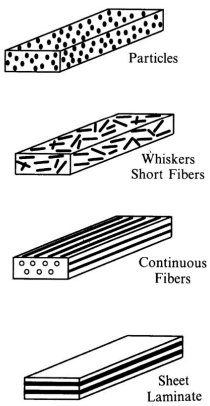
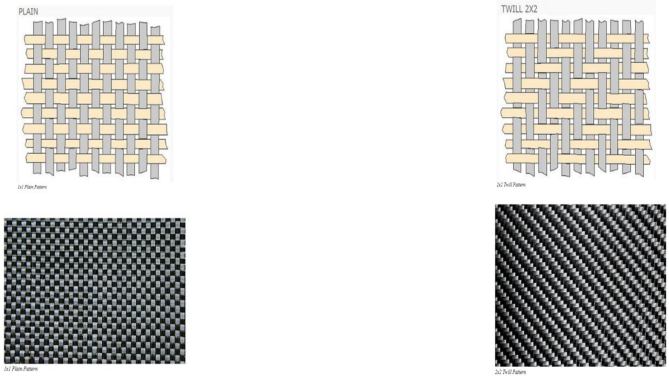


Figure 28: Long and short fibers
[43]

Experienced stress

The weave pattern is another factor that must be reviewed, this only applies to the use of continuous fibres. The act of weaving fibres together assists in encouraging a strong bond to accomodate for the high applied stresses during motion and this allows the material to perform in such a way to hold it's structure together without falling apart. There are several variations of patterns that can be chosen from, the choice is primarily related to stresses experienced in certain directions during application. Below are the two weave patterns most commonly used for many industries that require advanced composites. The Twill 2x2 is likely to be of high preference as the ease of handling of the fabric allow for complex geometries to be moulded, this applies to very sharp corners and intricate designs of the prototype.[44] The 1x1 Plain weave pattern involve bundles of fibre running across the longitudinal and transverse direction over and under every perpendicular bundle that results in a checkered design pattern. The 2x2 Twill has bundles of fibre running over and under every 2 bundles it passes, this pattern generally has slightly better properties due to less constraints and stress concentration near the contact regions between fibres, often known as 'crimping'. Compared with Plain weave, Twill is often known to be superior due to less crimping for the fewer contact regions between longitudinal and transverse bundles, consequently, Twill weave tends to be higher in strength. This is because fibres prefer to pull straight under tension, if there are crimps this means it is unable to do so without introducing large stresses onto the fibres.[45]

The diagram below demonstrates the cross-sectional view of the laminate where fibres are running perpendicular to one another. Crimping can be seen at the contact regions between the two directions and it is particularly affected worse when the fibre cross-section are of circular shape which, therefore has a higher crimp angle.



(a) Plain weave pattern (b) Twill weave pattern

Figure 29: Types of weave patterns

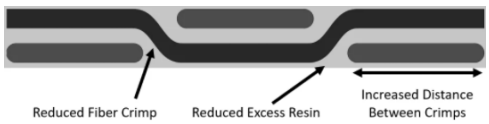


Figure 30: Cross-sectional view of the laminate diagram
[47]

Weight

As previously mentioned, we are aiming to have the lowest weight possible and still being able to achieve strength in the required directions. The diagram below suggests an example of the layup of different angled plies, this will still be looked into to determine the correct directions according to the different stresses in certain directions. Modelling and simulations of the chassis is necessary to determine the best configuration for the prototype. Considering both factors in terms of weight and stresses experienced, we want to optimize the strength and stiffness properties whilst still maintaining a light weight profile. The number of layers dictate the weight considerably and must be determined in order to obtain the desired configuration. As the number of layers increase with costs, the layers are kept on the lower margin, typically we are looking at between 5-10 layers but this is highly dependant on both funding and properties required. The directions that experience the least stresses do not need as much reinforcements in those directions which reduces the total weight and increasing mobility.[48]

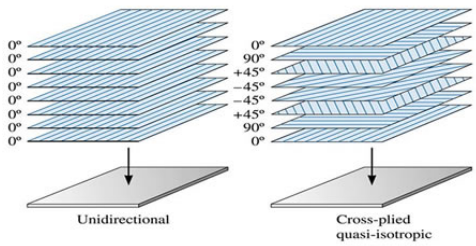


Figure 31: Layup of different angled plies
[49]

9.2 Chassis shape

Different shapes and sizes have been considered for the design of the chassis, which on a later stage will be tested and decided based on the results obtained: a ladder frame, a box frame, a fuselage structure or a flat plate.

Ladder Frame

A ladder frame consists of two long beams that are joined using two smaller beams to create a rectangular shape. The most major advantage of this arrangement is the ability to naturally withstand greater axial loads created through the pusher and during braking. The design and manufacturing are comparatively easier with additional ease of access to sub-assemblies. Drawbacks include the severe lack of torsional stiffness without the addition of extra beams (that subsequently increase weight) and the potential use of carbon fibre inherently complicating the attachment of the sub-assemblies.

Box Frame

Large beams run the length of the Pod, which attach to the chassis, and smaller beams connect these in order to provide structural support in areas of high load. This allows for good torsional strength however begins to add weight in comparison to the ladder frame. Most road cars use an iteration of the box frame, with many high-speed cars favouring a tub chassis.

Fuselage Structure

The fuselage structure involves arches that are spaced along the body of the chassis, connected by beams that run the length of the body. A direct connection is also provided to the outer shell of the Pod Adequate strength is demonstrated in the axial, torsional and radial directions with the largest issue being the added weight. This design is favourable in situations where a cabin is required for transport.

Flat plate

Using a simple flat plate for the chassis would make the manufacturing stage of it easier, and the application of materials like carbon fibers really effective. Despite this, it would as well imply an increase in weight for no apparent advantages from having these extra area of material, area which we predict that with any of the designs above would not be critical zones.[50]

9.3 Simulations

The initial plan was to perform different loading cases FE simulations on some initial chassis CAD designs but the conclusion that it was better to continue with research and defining our different procedure options to later in the design process conduct them was reached.

10 Projected Timeline

Having successfully completed the intensive research phase of the project, the team will proceed to design and prototype the Pod.

Based on the research-led decisions taken by each sub-team, the main target for the next 3 months will be to iterate the design of each component and rapidly prototype them using the society 3D printer, the Hackspace and other resources kindly provided by our Sponsors.

By the end of April, the objective is to have reached a functional design so that the first prototype can be manufactured before Summer 2021. Testing will follow and the team decides on what to change and improve on. During the summer term of 2022, the team will build the final ICLoop Pod to compete in the 2022 SpaceX Hyperloop Pod Competition.

Appendix

A SpaceX Pod competition track specifications

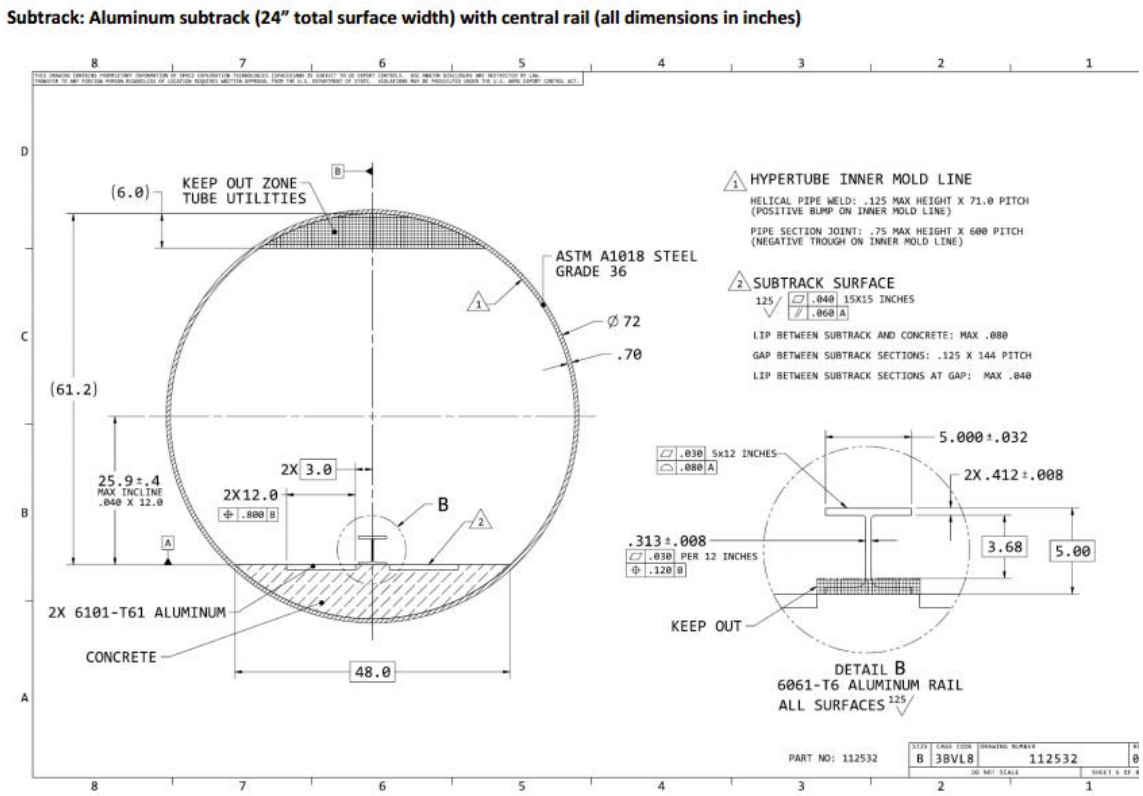
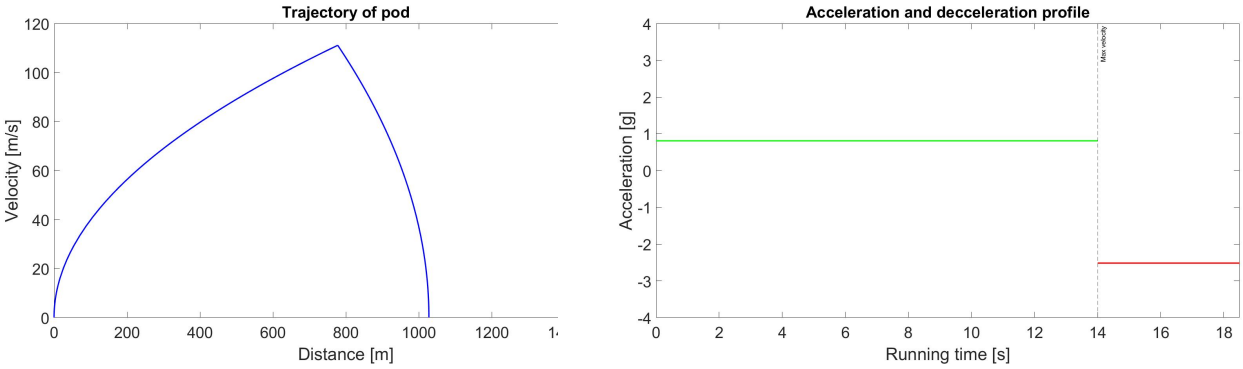


Figure A.1: SpaceX Track specifications

B Pod trajectory and kinematic data

This section describes the trajectory the Pod will follow, under the competition specifications given by SpaceX. This trajectory data was obtained for a 1 mile track with assumed acceleration and deceleration times and a target velocity of 400 km/h, allowing for over 500m of track as a safety margin. This kinematic dataset is only a first iteration and may be updated in further iterations. Also, the safety margin can be reduced in order to achieve higher top speeds or to travel at a fixed cruising speed.



(a) Pod trajectory

(b) Acceleration and deceleration profiles

Figure B.1: Pod trajectory profiles for first iteration

Table 1: Pod kinematic data for first iteration

	Kinematic data
Pod acceleration [ms^{-2}]	7.94
Braking acceleration [ms^{-2}]	-24.69
Total acceleration distance [m]	777.78
Total braking distance [m]	250.00
Total run distance [m]	1027.78
Extra distance [m]	572.22
Acceleration time [s]	14.00
Deceleration time [s]	4.50
Total run time [s]	18.50
Force to accelerate [N]	1984.13
Force to brake [N]	-6172.84

C CFD Parameters

Table 2: Parameters used in Star-CCM+

Characteristic length [m]	1
Freestream flow velocity [m/s]	100
Reference pressure [Pa]	1000
Turbulence Velocity scale [m/s]	1
Turbulence Specification	Intensity + Length scale
Turbulence Length scale [m]	0.19
Static Temperature [K]	293.15
Dynamic viscosity [Pa-s]	1.83e-5

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